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Operation regimes of spinal circuits controlling locomotion and role of supraspinal drives and sensory feedback

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Ilya A Rybak, Natalia A Shevtsova, Sergey N Markin, Boris I Prilutsky, Alain Frigon

Department of Neurobiology and Anatomy, College of Medicine, Drexel University, Philadelphia, Pennsylvania 19129, USA • School of Biological Sciences, Georgia Institute of Technology, Atlanta, Georgia 30332, USA • Department of Pharmacology-Physiology, Faculty of Medicine and Health Sciences, Centre de Recherche du CHUS, Université de Sherbrooke, Sherbrooke, Quebec J1H 5N4, Canada

Abstract

Locomotion in mammals is directly controlled by the spinal neuronal network, operating under the control of supraspinal signals and somatosensory feedback that interact with each other. However, the functional architecture of the spinal locomotor network, its operation regimes, and the role of supraspinal and sensory feedback in different locomotor behaviors, including at different speeds, remain unclear. We developed a computational model of spinal locomotor circuits receiving supraspinal drives and limb sensory feedback that could reproduce multiple experimental data obtained in intact and spinal-transected cats during tied-belt and split-belt treadmill locomotion. We provide evidence that the spinal locomotor network operates in different regimes depending on locomotor speed. In an intact system, at slow speeds (< 0.4 m/s), the spinal network operates in a non-oscillating state-machine regime and requires sensory feedback or external inputs for phase transitions. Removing sensory feedback related to limb extension prevents locomotor oscillations at slow speeds. With increasing speed and supraspinal drives, the spinal network switches to a flexor-driven oscillatory regime and then to a classical half-center regime. Following spinal transection, the model predicts that the spinal network can only operate in the state-machine regime. Our results suggest that the spinal network operates in different regimes for slow exploratory and fast escape locomotor behaviors, making use of different control mechanisms.

eLife assessment

This **fundamental** state-of-the-art modeling study explores neural mechanisms underlying walking control in cats, demonstrating the probability of three different states of operation of the spinal circuitry generating locomotion at different speeds. The authors' biophysical modeling sufficiently reproduces and provides explanations for experimental data on how the locomotor cycle and phase durations depend on treadmill walking speed and points to new principles of circuit functional architecture and operating regimes underlying how spinal circuits interact with supraspinal signals and limb sensory feedback signals to produce different locomotor behaviors at different speeds, which are major unresolved problems in the field. The modeling evidence is **compelling**, especially in advancing our understanding of locomotion control mechanisms and will interest neuroscientists studying the neural control of movement.

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Introduction

Locomotion is generated and controlled by three main neural components that interact dynamically [1-6]. The spinal network, including circuits of the central pattern generator (CPG), generates the basic locomotor pattern, characterized by alternation of flexor and extensor activity in each limb and coordination of activities related to left and right limbs. Supraspinal structures initiate and terminate locomotion and control voluntary aspects of locomotion. Somatosensory feedback from primary afferents originating in muscle, joint, and skin mechanoreceptors provides information on the state of the musculoskeletal system and the external environment. Although we know that the spinal network generates the basic locomotor pattern, its functional architecture, operating regimes, and the way it interacts with supraspinal signals and somatosensory feedback to produce different locomotor behaviors, including at different speeds, remain poorly understood.

In terrestrial locomotion, the step cycle of each limb consists of two main phases, swing and stance, that correspond primarily to the activity of limb flexors and extensors, respectively. A main role of corresponding spinal rhythm-generating circuits is to establish the frequency of oscillations, or cycle duration, and the durations of flexor and extensor phases [7]. In mammals, including humans, increasing walking speed leads to a decrease in cycle duration mostly because of a decrease in stance/extensor phase duration while swing/flexor phase duration remains relatively unchanged (reviewed in [8–10]). This is observed on a treadmill in intact animals and also following complete spinal thoracic transection (spinal animals). In intact animals, supraspinal signals interact with spinal networks and sensory feedback from the limbs, whereas in spinal animals, supraspinal signals are absent.

As an experimental basis of this study, we used data previously obtained in intact and spinal cats stepping on a treadmill with two independently controlled belts in tied-belt (equal speeds of left and right belts) and split-belt (different speeds of left and right belts) conditions [11–13]. Changes in cycle and phase durations during tied-belt and split-belt locomotion with increasing speed and left-right speed differences were similar in intact and spinal cats over a range of moderate speeds (0.4 – 1.0 m/s). This is somewhat surprising because intact and spinal cats rely on different control mechanisms. Intact cats walking freely on a treadmill engage vision for orientation in space and their supraspinal structures process visual information and send inputs to the spinal cord to control locomotion on a treadmill that maintains a fixed position of the animal relative to the external space. Spinal cats, whose position on the treadmill relative to the external space is fixed by an experimenter, can only use sensory feedback from the hindlimbs to adjust locomotion to the treadmill speed. An interesting additional observation here is that intact cats cannot consistently perform treadmill locomotion at very slow speeds (below 0.3-0.4 m/s), whereas spinal cats have no problem walking at such slow speeds [12,14]. This is evidently context dependent and specific for treadmill locomotion as cats, humans, and other animals can voluntarily decide to perform consistent overground locomotion at slow speeds.

To investigate the organization and operation of spinal circuits and how different mechanisms interact to control locomotion, we developed a tractable computational model of these circuits operating under the control of both supraspinal drives and sensory feedback. Our model followed the commonly accepted concept that the spinal cord contains CPG circuits that can intrinsically generate locomotor-like oscillations without rhythmic external inputs [4–7,15–20]. The concept of spinal mechanisms that intrinsically generate the basic locomotor pattern (CPG prototype) was initially proposed by Thomas Graham Brown [15,16] in opposition to the previously prevailing viewpoint of Maurice Philippson and Charles Sherrington [21–23] that locomotion is generated through a chain of reflexes, i.e., critically depends on limb sensory feedback (reviewed in [24,25]).

The present model is based on our previous models [26–32] and contains two rhythm generators (RGs) that control the left and right hindlimbs and interact through a series of commissural pathways. The RGs receive supraspinal drive, and limb sensory feedback during the extension phase. Each RG consists of flexor and extensor half-centers operating as conditional bursters (i.e., capable of intrinsically generating rhythmic bursting in certain conditions) and inhibiting each other. However, each RG is considered not as a simple neuronal oscillator, but as a neural structure with a dual function: as a state machine, defining operation in each state/locomotor phase independent of the phase-transition mechanisms, and as an actual oscillator, defining the mechanisms of state/phase transitions. A state machine, also called a finite-state machine, is a behavior model that can operate in only one of a finite number of states at any given time and may change its state (performing a *state transition*) in response to an external input [33,34]. We propose and show that, depending on conditions, each RG can operate in three different regimes: a non-oscillating *state-machine* regime, and in a *flexor-driven* and a *classical half-center* oscillatory regimes. In the full model, the generated locomotor behavior depends on the excitatory supraspinal drives to the RGs and on limb sensory feedback. The gain of limb sensory feedback in the intact model is suppressed by supraspinal drive via presynaptic inhibition [35–37] and is released from inhibition following spinal transection.

Our model reproduces and proposes explanations for experimental data, including the dependence of main locomotor characteristics on treadmill speed in intact and spinal cats. Particularly, based on our simulations, we suggest that locomotion in intact cats at low speeds and in spinal cats at any speed is mainly controlled by limb sensory feedback, consistent with Philippson's and Sherrington's view [21–23], whereas locomotion in intact cats at higher/moderate speeds is primarily controlled by the intrinsic oscillatory activity within the spinal network, supporting Brown's concept [15,16].

Results

Cycle and phase durations during tied-belt and split-belt locomotion in intact and spinal cats

Tied-belt locomotion

Our previous studies in intact (**Fig. 1A**) and spinal (**Fig. 1B**) cats showed a decrease in cycle duration with increasing speed due to a shortening of the stance phase with a relatively constant swing phase duration during tied-belt locomotion [11–13]. This agrees with other studies in mammals [8–10,38]. Note that most studies of treadmill locomotion in intact cats have been performed at treadmill speeds at or above 0.4 m/s (as in **Fig. 1A**). When placed on a treadmill moving at lower speed (from 0.1 m/s to 0.3 m/s), these cats demonstrate inconsistent stepping by making a few steps alternating with periods of stopping and/or sitting. Thus, when comparing intact and spinal cats (**Fig. 1C**), we highlight the following observations: (1) Intact cats do not step consistently during quadrupedal tied-belt locomotion on a treadmill if the speed is at or below 0.3 m/s, whereas spinal cats have no problem performing hindlimb locomotion from 0.1 - 0.3 m/s. (2) Speed- dependent changes in swing and stance phase durations from 0.4 to 1.0 m/s are qualitatively similar in intact and spinal cats, but stance duration in intact cats is usually a little longer. (3) With increasing treadmill speed, the duty factor (ratio between stance and cycle durations) in intact and spinal cats decreases towards 0.5 (equal swing and stance proportion). In spinal cats, the duty factor reaches 0.5 at approximately 0.8 - 0.9 m/s.

Split-belt locomotion

We used data from our previous studies in intact (**Fig. 2A**) and spinal (**Fig. 2B**) cats during split- belt locomotion, where the left (slow) hindlimb stepped at 0.4 m/s and right (fast) hindlimb stepped from 0.5 to 1.0 m/s [11–13]. When comparing intact and spinal cats, we highlight the following observations: (1) Increasing the speed of the fast belt leads to a decrease in cycle duration in intact and spinal cats, but the slow and fast hindlimbs maintain equal cycle duration; (2) In the slow hindlimb, increasing the speed of the fast belt leads to a small decrease in stance and swing durations in intact cats and in swing duration of spinal cats; (3) In the fast hindlimb, increasing the speed of the fast belt produces a decrease in the stance duration and an increase in swing duration in both intact and spinal cats. The duty factor reaches 0.5 at 0.9 m/s and 0.8 m/s in intact and spinal cats, respectively, and goes below this value at 0.9 – 1.0 m/s in spinal cats, where swing duration occupies a greater proportion of the cycle.

Conceptual framework and model description

Basic model architecture

In this study, we fully accept the concept that the mammalian spinal cord contains neural circuits (i.e. CPG) that can (in certain conditions) intrinsically (i.e. without rhythmic or patterned external inputs) generate the complex coordinated pattern of locomotor activity [4,6,7,15,17,18,39,40]. Following our previous computational models [26–28,30–32], we assume that the circuitry of the spinal locomotor CPG includes (a) rhythm-generating (RG) circuits that control states and rhythmic activity of each limb, and (b) rhythm-coordinating circuits that mediate neuronal interactions between the RG circuits and define phase relationships between their activities (coupling, synchronization, alternation). Also, following the classical view, we assume that each RG consists of two half- centers that mutually inhibit each other and define the two major states/phases of the RG, the flexor and extensors phases, in which the corresponding sets of limb muscles are activated [6,7,15,39].

Modeling of a single rhythm generator (RG) and its operation regimes

The ability of a RG to generate rhythmic activity can be based on the intrinsic rhythmic bursting properties of one or both half-centers or can critically depend on the inhibitory interactions between half-centers, so that each half-center cannot intrinsically generate rhythmic bursting, as

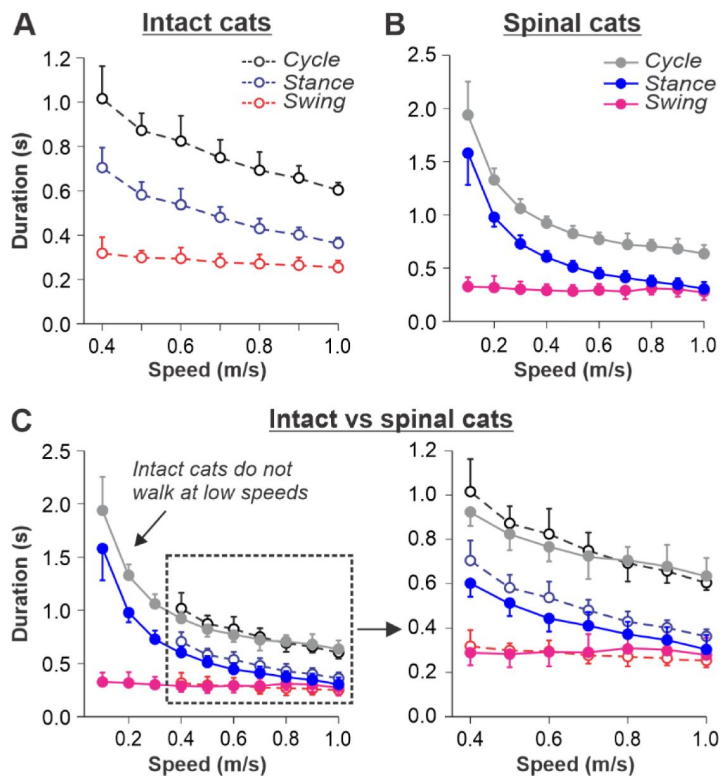

Fig. 1.

Locomotion of intact and spinal cats on a tied-belt treadmill.

A and **B**. Step cycle, stance and swing phase durations for the right hindlimb during tied-belt treadmill locomotion of intact (**A**, from Frigon et al., 2015; Latash et al., 2020) and spinal (**B**, from Frigon et al., 2017; Latash et al., 2020) cats with an increasing treadmill speed. Data were obtained from 6-15 cycles in seven intact and six spinal cats (one cat was studied in both states). Modified from **Fig. 3C, D** of Latash et al., 2020, under the license CC-BY-4. **C**. Superimposed curves from **A** and **B** to highlight differences.

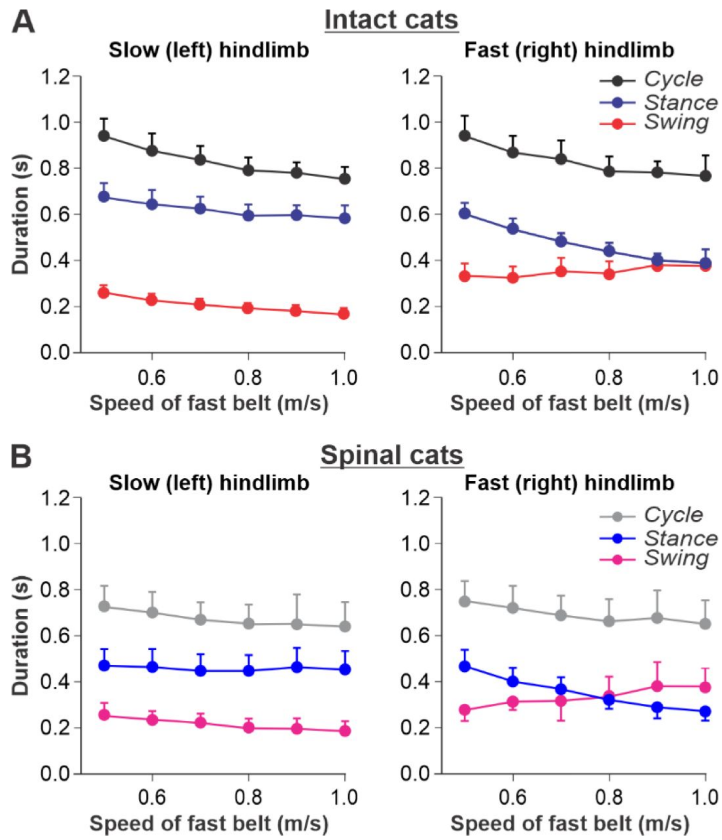

Fig. 2.

Locomotion of intact and spinal cats on a split-belt treadmill.

A. Step cycle, stance and swing phase durations for the left (slow) and right (fast) hindlimbs during split-belt treadmill locomotion of intact cats (from Frigon et al., 2015; Latash et al., 2020). **B.** Changes in the same characteristics for the left (slow) and right (fast) hindlimbs during split-belt treadmill locomotion in spinal cats (from Frigon et al., 2017; Latash et al., 2020). In both series of experiments the left (slow) hindlimb was stepping at 0.4 m/s while the right (fast) hindlimb stepped with speeds from 0.5 to 1.0 m/s with 0.1 m/s increments. Data were obtained from 6–15 cycles in seven intact and six spinal cats (one cat was studied in both states). Modified from **Fig. 6A,B** of Latash et al., 2020, under the license CC-BY-4.

in the classical half-center concept [15,16]. However, in the isolated mouse spinal cord using optogenetic stimulation, rhythmic activity can be induced independently in flexor- and extensor-related spinal circuits [41], suggesting that both flexor and extensor half-centers can, in certain conditions, generate independent rhythmic bursting activity (i.e. operate as conditional bursters). Previous mathematical models of neurons defined and analyzed the conditions allowing these neurons to generate rhythmic bursting based on nonlinear voltage-dependent properties of different ionic channels [42–46]. Several models that described bursting activity in neurons of the medullary pre-Bötzinger complex for respiration or in the spinal cord for locomotion, suggested that this activity is based on a persistent (slowly-inactivating) sodium current, I_{NaP}[26,47–52]. We implemented a similar I_{NaP} -dependent mechanism for conditional bursting in our RG half-center model (see Methods).

Figure 3 illustrates the behavior of a neuron model, describing a single conditional burster (**Fig. 3A**) with the output representing changes in spike frequency (see Methods). At low excitatory drive, the neuronal output is equal to zero ("*silence*"), but when the drive exceeds some threshold, the model switches to a "*bursting*" regime, during which the frequency of bursts increases with increasing drive (**Fig. 3B**). At a certain drive, the model switches to sustained "*tonic*" activity.

Let us now consider a simple half-center RG consisting of two conditional bursters, the flexor half-center (F), receiving an increasing Drive-F, and the extensor half-center (E), receiving a constant Drive-E that maintains it in a state of tonic activity if uncoupled. The two half-centers inhibit each other through inhibitory neurons, InF and InE (**Fig. 3C**). At low Drive-F there are no oscillations, and the F half-center remains silent, although it could start oscillating if it would not be strongly inhibited by the E half-center. The E half-center shows tonic activity because of the high value of Drive-E. We call this regime of RG operation a *state-machine* regime (**Fig. 3D**, left part of the graph). In this regime, the RG maintains a state of extension, until an external signal, sufficiently strong, activates the F half-center or inhibits the E half-center (green arrows in **Fig. 3C**) to release the F half-center from inhibition, allowing it to generate intrinsic bursts, switching the model to a flexion state. Increasing the excitatory Drive-F releases the F half-center from the E half-center's inhibition (acting via the InE neuron), switching the RG to a bursting regime. Note that in this regime, the E half-center also exhibits bursting activity in alternation with the F half-center due to rhythmic inhibition from the flexor-half center via the InF neuron, although the E half-center itself, if uncoupled, operates in the tonic mode. We call this regime a *flexor-driven* regime (**Fig. 3D**, middle part of the graph). Similar to the intrinsic bursting regime in an isolated conditional burster (**Fig. 3B**), with an increase in Drive-F, the bursting frequency of the RG is increasing (and the oscillation period is decreasing) mostly due to shortening of the extensor bursts with much less reduction in flexor burst duration. Further increasing the excitatory Drive-F leads to a transition of the RG to a *classical half-center* regime (**Fig. 3D**, right part of the graph), in which the half-centers cannot generate oscillations if uncoupled, and RG oscillations occur due to, and critically depend on, mutual inhibition between half-centers and an adaptive reduction of their responses. In this regime, with increasing Drive-F, the oscillation frequency (and period) remains almost unchanged, and flexor burst duration increases to compensate for decreasing extensor bursts. To summarize, increasing the excitatory drive to the flexor half-center in this simple RG model demonstrates a sequential transition from a *state-machine* regime to a *flexor-driven* regime, and then to a *classical half-center* regime. The sensitivity analysis of this model can be found in **Supplementary file 1** and **Fig. S1**.

Model of the spinal locomotor circuitry

The schematic of our model is shown in **Fig. 4**. The model incorporates neuronal circuits involved in the control of, and interactions between, two cat hindlimbs during locomotion on a treadmill. The spinal circuitry in the model includes two RGs (as described above) interacting via a series of commissural interneuronal (CIN) pathways mediated by different sets of genetically identified commissural ($V0_D$, V0V, and V3) and ipsilaterally projecting (V2a) interneurons, as well

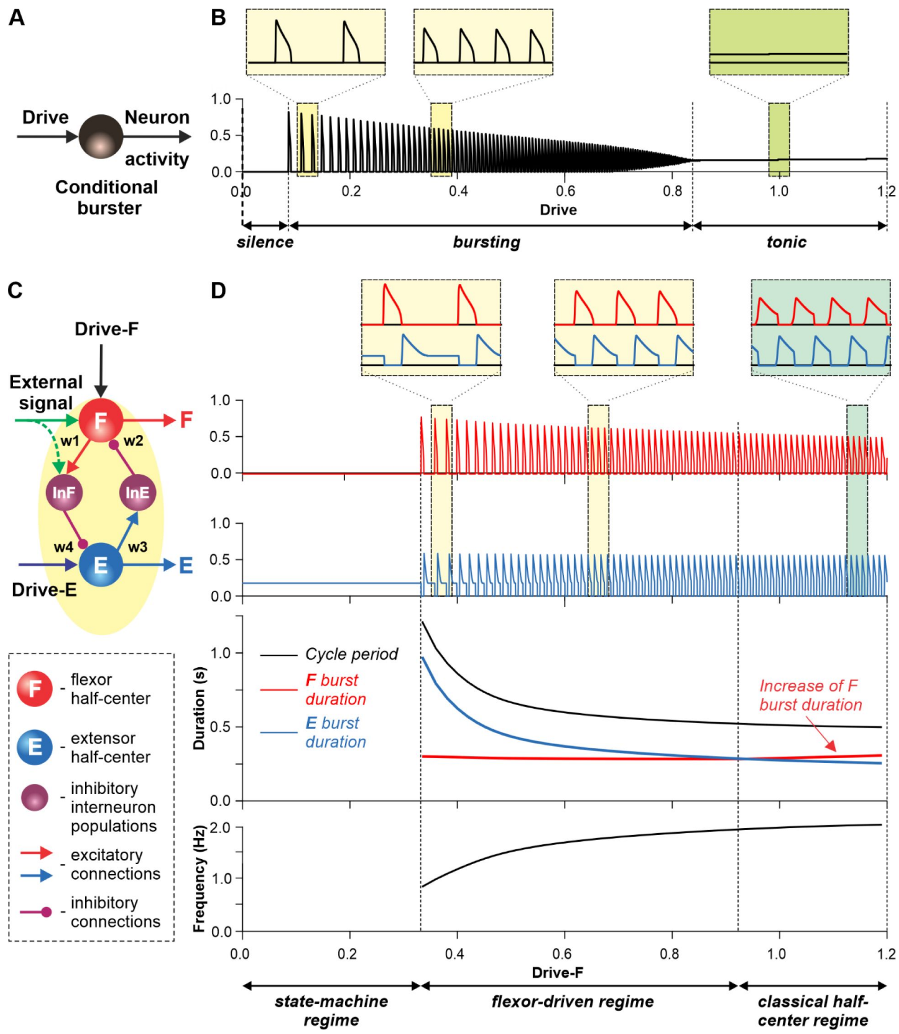

Fig. 3.

Modeling a single conditional burster and a half-center rhythm-generator.

A. The behavior of a single I_{NaP}-dependent conditional burster. **B.** Changes in the burster's output when the excitatory input (Drive) progressively increases from 0 to 1.2. With increasing Drive, the initial *silence* state (zero output) at low Drive values changes to an intrinsic *bursting* regime with burst frequency increasing with the Drive value (seen in two left insets), and then to a *tonic* activity (seen in right inset). **C.** Model of a simple half-center network (RG) consisting of two conditional bursters/half-centers inhibiting each other through additional inhibitory neurons, InF and InE. The flexor half-center (F) receives progressively increasing Drive-F, whereas the extensor half-center (E) receives a constant Drive-E keeping it in the regime of tonic activity if uncoupled. **D**. Model performance. At low Drive-F values, there are no oscillations in the system. This is a *state-machine* regime in which the RG maintains the state of extension, until an external (strong enough) signal arrives to activate the F half-center or to inhibit the E half-center (see green arrows) to release the F half-center from E inhibition allowing it to generate an intrinsic burst. Further increasing the Drive-F releases the F half-center from E inhibition and switches the RG to the bursting regime (see two insets in the middle). In this regime, the E half-center also exhibits bursting activity (alternating with F bursts) due to rhythmic inhibition from the F half-center. This is a *flexor-driven* regime. In this regime, with an increase in Drive-F, the bursting frequency of the RG is increasing (and the oscillation period is decreasing) due to shortening of the extensor bursts with much less reduction in the duration of flexor bursts (see bottom curves and two left insets). Further increasing the excitatory Drive-F leads to a transition of RG operation to a *classical half-center* oscillatory regime, in which none of the half-centers can generate oscillations if uncoupled, and the RG oscillations occur due to mutual inhibition between the half-centers and adaptive properties of their responses. Also in this regime, with an increase of Drive-F, the period of oscillations remains almost unchanged, and the duration of flexor bursts is increasing partly to compensate for the shortening of extensor bursts, which is opposite to the flexor-driven regime (see bottom curves and right inset).

as some hypothetical inhibitory interneurons (Ini). The organization of these intraspinal interactions was directly drawn from our earlier models [26–32] that were explicitly or implicitly based on the results of molecular/genetic studies of locomotion in mice or were proposed to explain and reproduce multiple aspects of the neural control of locomotion in these studies. Specifically, $V0_D$ and V2a- $V0_V$-Ini pathways secure left-right alternation of RG oscillations during walking and trotting at slow and higher locomotor speeds, respectively [26,27,53]. The V3 CINs contribute to left-right synchronization of RG oscillations during gallop and bound [26–28,30–32].

The proposed spinal network connectome allowed the previous models to reproduce and explain multiple experimental phenomena observed during fictive and real locomotion, including the speed-dependent expression of different gaits and its supraspinal control [26–28,30–32]. The molecular/genetic types of neurons in this connectome and their known and suggested interactions were based on studies of intact and mutant mice obtained by optogenetic methods, most of which are not available for studies in cats. We assume that the neural connectome in the spinal cord of mammals is evolutionarily conserved and findings from studies in mice can be used for understanding the neural control of locomotion in other quadrupedal mammals, such as cats.

Control of spinal locomotor circuits by supraspinal drives and sensory feedback and interactions between them

In our intact model, the frequency of locomotor oscillations (and the speed of locomotion) is primarily controlled by excitatory supraspinal drives to both RGs, particularly to the flexor half-centers (**Fig. 4A**). These drives simulate the major descending brainstem pathways to spinal neural circuits. Our previous models also suggested that some supraspinal drives activate/inhibit CINs to modify interlimb coordination and perform gait transitions from left-right alternating (walk, trot) to left-right synchronized (gallop, bound) [26–28,30–32]. In the present model, this feature is implemented by supraspinal excitatory signals to V3 CINs (**Fig. 4A**).

As described for the single RG model (**Fig. 3B**), both left and right RGs start to intrinsically generate rhythmic locomotor activity when the value of supraspinal drives to the flexor half-centers exceeds some threshold. Below this threshold, the RGs can only operate in a state-machine regime that requires an additional excitatory input to the flexor half-centers to initiate flexion. Such additional trigger signals can come from supraspinal structures, such as the motor cortex, or from limb sensory feedback. In the spinal-transected model that lacks supraspinal drives (**Fig. 4B**), and in the intact model with low drives necessary for a very slow locomotion, both RGs (specifically, their F half-centers) do not receive sufficient excitation and can only operate in a state-machine regime (see **Fig. 3D**), in which the extension to flexion transition requires additional signals that can be provided by limb sensory feedback.

It is well known that the operation of the spinal network during locomotion is also controlled by inputs from primary afferents originating in muscle spindles (groups Ia and II), Golgi tendon organs (group Ib) and different skin mechanoreceptors (reviewed in [2]). Limb sensory feedback controls phase durations and transitions and reinforces extensor activity during stance. We incorporated two types of limb sensory feedback (SF) in our model in a simplified version, both operating during the extensor phase of each limb.

The first feedback, SF-E1, represents an increase of spindle afferent activity from hindlimb flexor muscles as they are stretched with limb extension (an increase of hip angle) during stance [54], which increases with increasing treadmill speed. This feedback directly activates the ipsilateral F half-center (**Fig. 4**) and promotes the transition from extension to flexion (simulating lift-off and the stance-to-swing transition). The same feedback acting in the model through the V3-E CIN and InE neurons inhibits the contralateral F half-center (**Fig. 4**) and promotes the transition from flexion to extension in the contralateral RG. The critical role of hip flexor stretch-related feedback

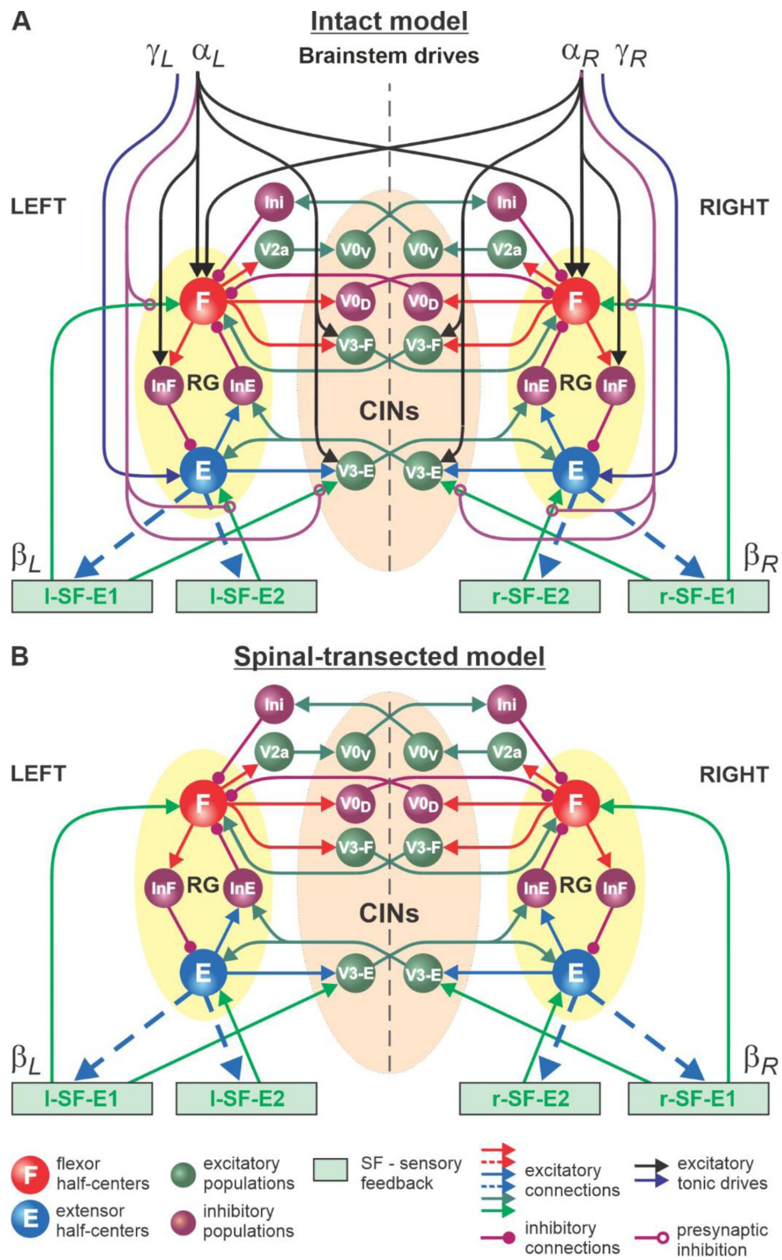

Fig. 4.

Model of spinal circuits controlling treadmill locomotion.

A. Model of the intact system ("intact model"). The model includes two bilaterally located (left and right) RGs (each is similar to that shown in **Fig. 3B**) coupled by (interacting via) several commissural pathways mediated by genetically identified commissural ($V0_D$, V0V, and V3) and ipsilaterally projecting excitatory (V2a) and inhibitory neurons (see text for details). Left and right excitatory supraspinal drives ($\alpha_L$and α_R) provide activation for the flexor half-centers (F) of the RGs (ipsi- and contralaterally) and some interneuron populations in the model, as well as for the extensor half-centers (E) (γ_L and γ_R ipsilaterally). Two types of feedback (SF-E1 and SF-E2) operating during ipsilateral extension affect (excite) respectively the ipsilateral F and E half-centers, and through V3-E neurons affect contralateral RGs. The SF-E1 feedback depends on the speed of the ipsilateral "belt" (β_L or β_R) and contributes to extension-to-flexion transition on the ipsilateral side. The SF-E2 feedback activates ipsilateral E half-center and contributes to "weight support" on the ipsilateral side. The ipsilateral excitatory drives (α_L and α_R) suppress (reduce) the effects of all ipsilateral feedback inputs by presynaptic inhibition. **B.** Model of spinal-transected system. All supraspinal drives (and their suppression of sensory feedback) are eliminated from the schematic shown in **A.**

for triggering the stance-to-swing (or extension-to-flexion) transition has been confirmed in many studies in cats during real and fictive locomotion [2,55–59]. The role of V3 CINs in transmitting primary afferent activity to the contralateral side was supported by a recent mouse study [60].

The second feedback we incorporated in our model, SF-E2, simulates in a simplified form the involvement of force-dependent Ib positive feedback from limb extensor muscles to the ipsilateral extensor half-center, reinforcing extensor activity and weight support during stance (**Fig. 4**). This effect and the role of Ib feedback from extensor afferents has been demonstrated and described in many studies in cats during real and fictive locomotion [2,61–64].

An important feature of our model is how supraspinal drives and limb sensory feedback interact with each other and with spinal circuits. Specifically, limb sensory feedback from each limb receives presynaptic inhibition from the ipsilateral supraspinal drive (**Fig. 4A**), which reduces feedback gains in a speed-dependent manner, suppressing the influence of both feedback types. This interaction in the model reflects experimental data on presynaptic and/or direct inhibition of primary afferents by supraspinal signals [35–37,65]. In our model, removing supraspinal drives to simulate the effect of a complete spinal transection also eliminates presynaptic inhibition of all inputs from sensory feedback, increasing the influence of both types of sensory feedback on spinal CPG circuits (**Fig. 4B**).

Intact animals walking on a treadmill use visual cues and supraspinal signals to adjust their speed and maintain a fixed position relative to the external space [66]. In a simplified version (implemented in the model), the movement speed relative to the treadmill belt is mainly controlled by supraspinal drives to the left and right RGs (parameters α_L and α_R, see Methods) that define the locomotor oscillation frequency. These drives are automatically adjusted to the speed of the simulated treadmill (parameters β_L and β_R, see Methods). The frequency of oscillations generated by the RGs and the duty factor of the generated pattern are also affected by SF-E1 and SF-E2 acting during the extension phases of each RG, but they are progressively suppressed by supraspinal drives through presynaptic inhibition with increasing speed. Thus, the role of feedback in the control of locomotion decreases with an increase of locomotor frequency defined by the increasing supraspinal drives. In the spinal-transected model, the transition from extension to flexion is fully controlled by limb sensory feedback. The extensor phase duration and the period (and frequency) of oscillations are mainly defined by the rate of SF-E1 increase, which in turn is defined by the speed of the simulated treadmill (parameters β_L and β_R, see Methods).

Simulation of tied-belt locomotion in intact and spinal cats

We simulated an increase in treadmill speed in the tied-belt condition by the progressive increase of parameters β_L=β_R. As stated above, we assumed that intact cats voluntary adjust supraspinal drives to the left and right RGs to the treadmill speed to maintain a fixed position relative to the external space. Therefore, in the intact model, the parameters α_L=α_R that characterized supraspinal drives were adjusted to correspond to the parameters β_L=β_R characterizing treadmill speed. In intact (**Fig. 5A**) and spinal-transected (**Fig. 5B**) models, changes in cycle and phase durations were qualitatively similar to experimental data (compare with **Figs. 1A,B**). Specifically, the oscillation period (cycle duration) decreases with an increasing “speed” due to a shortening of the extensor phase with a relatively constant flexor phase duration, so that the duty factor approaches or reaches 0.5. Comparison of changes in cycle and phase durations of the intact and spinal-transected models demonstrates similar dynamics of changes (**Fig. 5C**), matching corresponding experimental data (**Fig. 1C**). It is important to note that in the intact model at “slow treadmill speeds” (low β_L=β_R) and the corresponding weak supraspinal drives (α_L=α_R, approximately at or below 0.4), both RGs are unable to intrinsically generate rhythmic activity and operate in a state-machine regime where switching from the extensor to the flexor phase is mainly controlled by SF-1 and SF-2. At higher values of “treadmill speeds” and corresponding supraspinal drives, locomotion begins to be mainly controlled by supraspinal drives to the RGs. In this case, the role of limb sensory feedback in the control of phase durations is reduced because of increased presynaptic inhibition by increasing supraspinal drives (see **Fig. 5C**). In contrast to the intact model, the transected model has no supraspinal drives to the RGs and can only operate in a state-machine regime at all treadmill speeds. Thus, the transition from extension to flexion and the duration of the extensor phase are entirely controlled by sensory feedback.

Simulation of split-belt locomotion in intact and spinal cats

For split-belt locomotion, we simulated the speed of the “slow” RG with a fix value of β_L=0.4 and the “fast” RG with β_R progressively increasing from 0.5 to 1.0. Again, based on our assumption above, the supraspinal drives to the left and right RGs in the intact model were adjusted to the corresponding speeds of the treadmill belts. We set a constant drive α_L=0.4 to the left RG and progressively increased the drive α_R to the right RG from 0.5 to 1.0. We show that in the intact model (**Fig. 6A**), despite differences in the “speeds” of left and right RGs, the left and right oscillation periods are equal, which corresponds to experimental data (**Fig. 2A**).

In the spinal-transected model (**Fig. 6B**), changes in cycle and phase durations are similar to those in the intact model and correspond to experimental data (**Fig. 2B**). The oscillation periods on the left and right sides are equal, and changes in cycle and phase durations on the left slow side do not change much with a progressive increase of β_R from 0.5 to 1.0. The flexor phase duration of the fast RG increases with increasing β_R, which compensates for the decrease of the extensor phase duration.

An increase of flexion phase duration on the fast side with an increase of speed on that side in the spinal-transected model is much steeper than in the intact model (compare **Fig. 6B** with **Fig. 6A**) which reproduces corresponding experimental data (see **Fig. 2B** vs. **Fig. 2A**). The mechanism of this increase in the spinal case (spinal-transected model) is the following. With the increase of speed, the SF-E1 starts accumulating a tonic component that continues acting during flexion. This accumulated tonic activity provides a direct excitation to the ipsilateral F half-center during flexion, which is compensated by inhibition from the contralateral SF-E1 that inhibits the ipsilateral F half-center during the flexor phase (via contralateral V3-E CIN and ipsilateral InE neuron, see **Fig. 4**). In tied-belt locomotion (**Fig. 5**), this contralateral inhibition acts in balance with the accumulated ipsilateral excitation from SF-E1 and limits excitation of each F half-center during flexion, keeping both RGs within the flexor-driven operating regime with relatively constant flexor phase durations (as in **Fig 3D**, middle part of the graph). In split-belt locomotion, this contralateral inhibition from the slow (left) side remains relatively weak (it is fixed at a value corresponding to the constant slow left-side speed and does not change with the increasing speed on the right fast side), whereas the accumulated excitation on the fast side continues increasing with the speed). Therefore, the F half-center on the fast side becomes overexcited and the RG on the fast side starts operating in the classical half-center regime (like in **Fig. 3D**, right part of the graph) with an increasing flexor phase duration as the speed on the fast side increases (**Fig. 6B**). A similar mechanism operates in the intact split-belt case (**Fig. 6A**), but the resultant increase of the flexor phase duration on the fast side is much weaker due to presynaptic inhibition of left and right SF-E1 feedback.

Effects of feedback removal on treadmill locomotion

We used our model to simulate the effects of removing limb sensory feedback. The results of these simulations allowed us to formulate modeling predictions that could be tested in future experiments, providing an additional validation to our model. We specifically focused on simulation of separate removal of SF-E1 feedback, activated with limb extension and involved in the stance-to-swing transition, or SF-E2 feedback involved in the generation of extensor activity and weight support during stance, as well as removal of both feedback types.

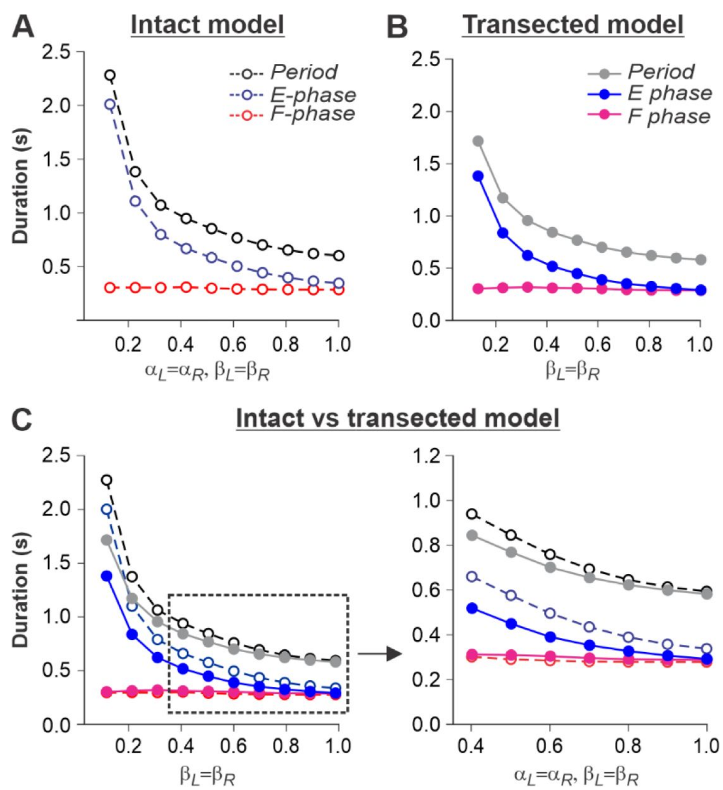

Fig. 5.

Simulation of locomotion on a tied-belt treadmill using intact and transected models.

A and **B**. Changes in the durations of locomotor period and flexor/stance and extensor/swing phases during simulated tied-belt locomotion using the intact (**Fig. 4A**) and transected (**Fig. 4B**) models with an increasing simulated treadmill speed. **C.** Superimposed curves from panels **A** and **B** to highlight differences.

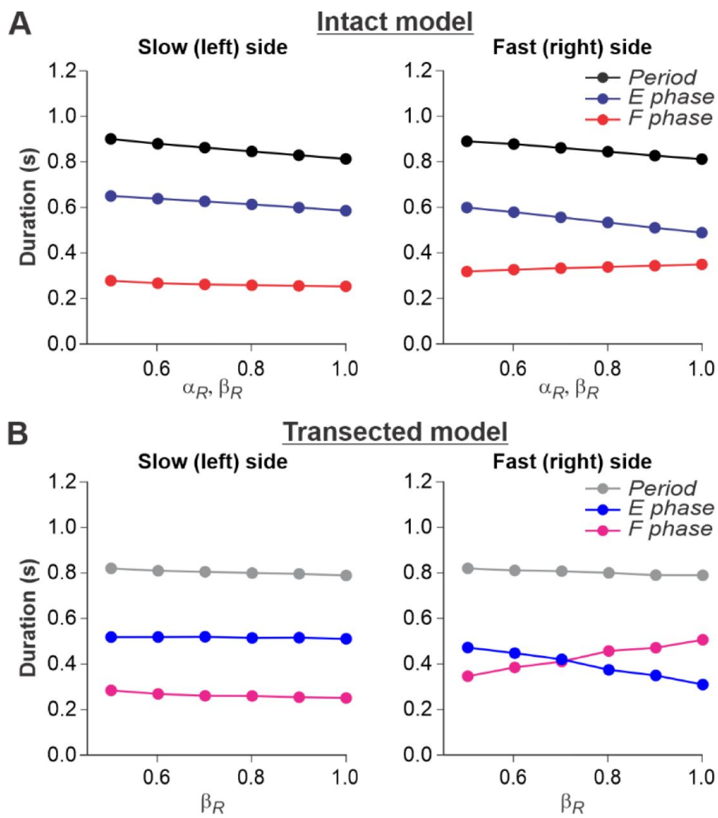

Fig. 6.

Simulation of locomotion on a split-belt treadmill using intact and transected models.

A. Changes in the durations of locomotor period and flexor/stance and extensor/swing phases for the left (slow) and right (fast) sides during split-belt treadmill locomotion using the intact model (**Fig. 4A**). **B.** Changes in the same characteristics for the left (slow) and right (fast) sides during simulation of split-belt treadmill locomotion using the transected model (Fig . 4B). In both cases, the speed of the simulated left (slow) belt was constant (β_L=0.4) while the speed of the simulated right belt (β_R) changed from 0.5 to 1.0 with 0.1 increments.

In the transected model, removal of SF-E1 (with or without SF-E2) prevented locomotion (not shown). Without supraspinal drive and SF-E1, both RGs could not switch to the flexor phase and generate locomotor activity. Following removal of SF-E2 in the transected model, sufficient extensor activity (necessary for “weight support”) could not be developed, leading to an abnormal locomotion with extremely short extensor phase durations (and oscillation periods), with an inability to adjust to treadmill speed (**Fig. 7**). In this case, locomotion could be recovered by adding an additional activation of extensor half-centers during stance phases (not shown).

The described above effects of removing limb sensory feedback on locomotion in the spinal-transected model were expected. However, what happens after removal of limb sensory feedback in the intact model, in which supraspinal drives are present? Removing SF-E1 in the intact model produced interesting results. It considerably increased the duration of the extensor phase (**Fig. 8A**). However, the model only demonstrated oscillatory activity starting at some moderate speed, when $\beta_L=\beta_R \geq 0.5$. This required the corresponding supraspinal drives to the RGs to be sufficient ($\alpha_L=\alpha_R \geq 0.5$) to operate in the flexor-driven regime and produce phase transitions without any contributions from SF-E1. In contrast, removing SF-E2 in the intact model reduced the activity of the extensor half-centers, shortening extensor phase durations but preserved oscillations in the full range of considered “speeds” (**Fig. 8B**). Therefore, the removal of SF-E2 produced an opposite effect on extensor phase durations compared with removing SF-E1. Finally, removing both feedback types in the intact model shifted the start of oscillatory activity to the left compared to removal of only SF-E1, allowing oscillations to start at $\beta_L=\beta_R \geq 0.35$ (**Fig. 8C**). This allows the prediction that below 0.35 m/s, cats with diminished limb sensory feedback can only perform locomotion with patterned step-by-step supraspinal signals to produce phase transitions. This can also explain why intact cats do not usually walk consistently on a treadmill with a speed at or below 0.3 m/s.

Discussion

Operation regimes of the spinal locomotor network

Despite decades of research, a commonly accepted definition of the spinal locomotor CPG has not been formulated, and different authors use this term in relation to different entities depending on the context. For instance, the term CPG has been used to designate spinal circuits controlling and coordinating rhythmic movements of all limbs, or controlling only rhythmic movements of a single limb, or even a single joint [4,6,7,15–19,40]. Here, we use the term CPG for the spinal circuitry controlling and coordinating all limbs, and the term RG (rhythm generator) for the relatively independent part of the CPG that controls rhythmic movements of a single limb. We consider the CPG as a group of RGs, each controlling a single limb and interacting with each other through commissural and/or propriospinal pathways or circuits [3,26–32].

Regardless of the exact name (RG or unit burst generator), a common view is that the function of the RG is to generate locomotor-like rhythmic bursting consisting of two major phases, flexion and extension. In each phase, the controlled limb operates in the corresponding functional state, defining the contraction and relaxation of specific sets of muscles. Thus, the RG is not just an oscillator, but rather a system with a dual function: it functions as a *state-machine* [67–69], defining the state and operation of the controlled biomechanical system in each phase independent of the exact transition mechanisms, and as *an oscillator* defining the mechanisms and timing of transitions between states. These transitions may be fully defined by internal properties of the RG and/or require a contribution from external inputs, such as supraspinal signals or somatosensory feedback, that control or adjust the timing of transitions.

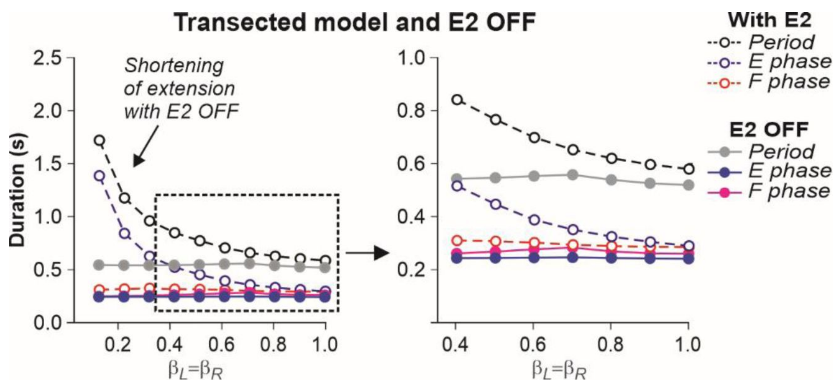

Fig. 7.

Simulation of the effect of removing SF-E2 feedback in the transected model during simulated tied-belt locomotion.

Changes in the durations of locomotor period and flexor/stance and extensor/swing phases during simulated tied-belt locomotion using the transected model (**Fig. 4B**) after removal of SF-E2 feedback.

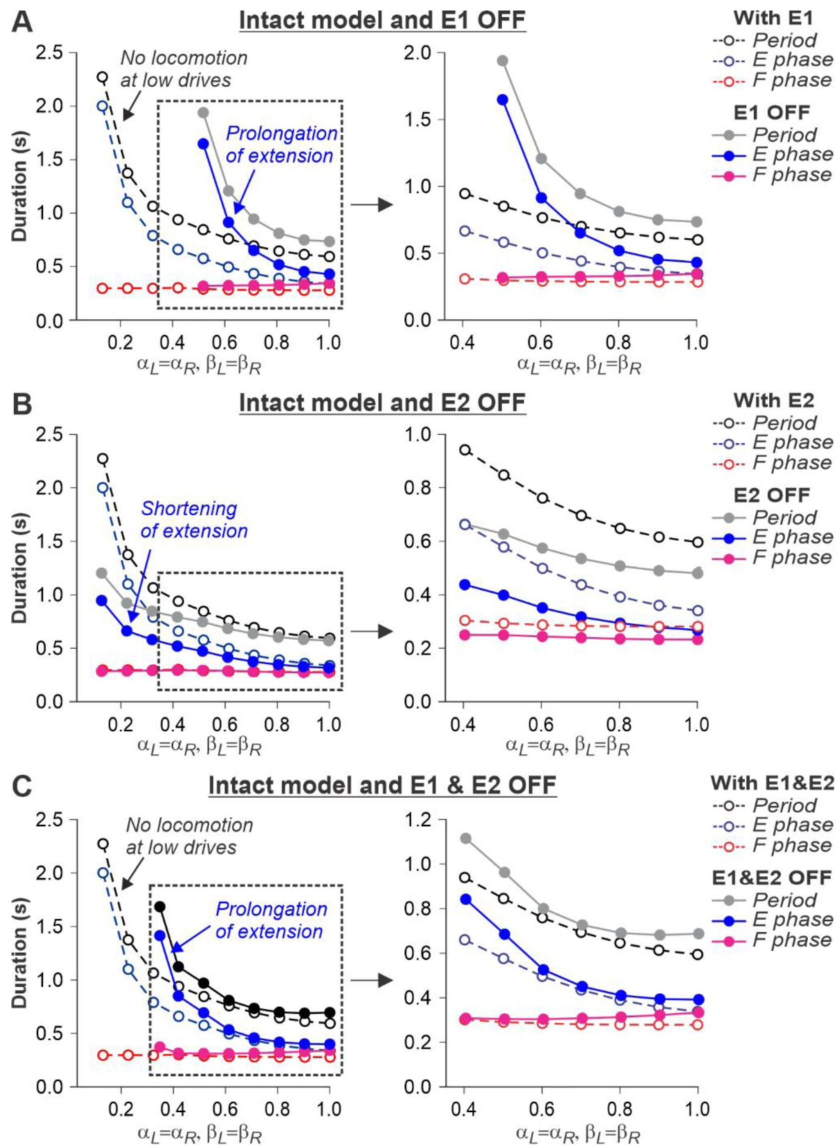

Fig. 8.

Simulation of the effect of removing SF-E1, or SF-E2, or both feedback types in the intact model during simulated tied-belt locomotion.

Changes in the durations of locomotor period and flexor/stance and extensor/swing phases during simulated tied-belt locomotion using intact model (**Fig. 4A**) with an increasing simulated treadmill speed after removal of only SF-E1 feedback (**A**), only SF-E2 feedback (**B)**, and both feedback types (**C**).

Here, we described a relatively simple model of a half-center RG. We identified three operation regimes: *state-machine*, *flexor-driven*, and *classical half-center*. In an intact system, at slow speeds (≤ 0.35 m/s), the spinal network operates in a state regime and requires external inputs for phase transitions, which can come from limb sensory feedback and/or volitional inputs (e.g. from the motor cortex). At higher speeds and greater supraspinal drives, the spinal network switches to a flexor-driven regime before transitioning to a classical half center regime at higher speeds/drives. Following spinal transection, the spinal network can only operate in the state regime and entirely depends on limb sensory feedback for phase transitions.

Our modeling results also resolve a potential contradiction between the classical half-center and flexor-driven concepts of spinal RG operation. According to the *classical half-center concept*, both flexor and extensor half-centers are necessary for generating rhythmic activity [7,15,16]. An alternative, *flexor-driven concept* was proposed by Pearson and Duysens ("swing generator model"), where the flexor half-center is intrinsically rhythmic, unlike the extensor half-center, which exhibits rhythmic activity due to rhythmic inhibition from the flexor half-center [70,71]. Here we follow our previous modeling studies [13,52] and show that these concepts do not contradict each other, but rather relate to different regimes of RG operation. With an increase of excitatory drive to an RG, the rhythmogenic mechanism changes from a flexor-driven to a classical half-center mechanism (**Fig. 3**). This transition explains the increase of the flexor phase duration of the fast hindlimb or RG during split-belt treadmill locomotion in intact and spinal cats (**Figs. 2 and 6**).

Intrinsic spinal rhythm-generating versus reflex-based mechanisms of locomotion

Following Maurice Philippson's view formulated from studies in spinal dogs [23], Charles Sherrington proposed that locomotion in decerebrate and spinal cats was generated by chains of reflexes [21,22]. Yet, Thomas Graham Brown clearly showed that an intrinsic spinal mechanism alone, without somatosensory feedback, could generate rhythmic alternating activity [15], which later became the CPG concept. Our results support both concepts, depending on locomotor speed and the state of the animal. Based on our simulations, we suggest that in spinal cats at any speed and in intact cats at low speeds, the spinal network operates in a state-machine regime and requires some sensory feedback to locomote, consistent with Philippson's/Sherrington's viewpoint. In contrast, at higher (moderate) speeds (≥ 0.4 m/s), when supraspinal drive is sufficient, the spinal network can intrinsically generate the basic locomotor activity controlling locomotion, which supports Brown's concept.

Our results also suggest an important difference in the control of slow exploratory and faster escape locomotion [72–75]. Based on our predictions, slow (conditionally exploratory) locomotion is not "automatic" but requires volitional (e.g. cortical) signals to trigger step-by-step phase transitions because the spinal network operates in a state-machine regime. In contrast, locomotion at moderate to high speeds (conditionally escape locomotion) occurs automatically under the control of spinal rhythm-generating circuits receiving supraspinal drives that define locomotor speed, unless voluntary modifications or precise stepping are required to navigate complex terrain[76,77].

We also used our model to simulate and predict the effects of removing limb sensory feedback on locomotion. As expected, the spinal-transected model failed to generate locomotion without SF-E1 or normal locomotion without SF-E2 because the spinal network operates in a state-machine regime (**Fig. 7**). The essential role of somatosensory feedback after spinal cord injury is well known [2,78–82]. However, in the intact model, limb sensory feedback, particularly from hip flexor afferents (SF-E1), is also required for locomotion and phase transitions at slow speeds to compensate for low supraspinal drives (**Fig. 8**).

Another important implication of our results relates to the recovery of walking in movement disorders, where the recovered pattern is generally very slow. For example, in people with spinal cord injury, the recovered walking pattern is generally less than 0.1 m/s and completely lacks automaticity [83–85]. Based on our predictions, because the spinal locomotor network operates in a state-machine regime at these slow speeds, subjects need volition, additional external drive (e.g., epidural spinal cord stimulation) or to make use of limb sensory feedback by changing their posture to perform phase transitions. Another paper commenting on observations in Parkinsonian patients also proposed that different CPG control systems operate at slow and fast speeds [86].

The concept of a spinal locomotor CPG in humans is strongly supported by a variety of experimental and clinical evidence [87–90]. In human walking, the basic output of the CPG, reflected in the EMG pattern of leg muscles, is maintained even at very slow speeds, albeit with differences in amplitudes and some bursting patterns from slow to fast walking speeds [91,92]. This is consistent with the idea that the same CPG provides the basic motor program from slow to fast speeds but that additional inputs or drives are required. The idea of a flexible arrangement of CPG circuits has also been proposed to explain the generation of locomotion and other rhythmic motor behaviors (e.g. scratching, paw shaking) in various species, such as cats, turtles, zebrafish and rats[93–98].

Model limitations and future directions

In this study, we developed a simplified model of the neural control of cat hindlimb locomotion in tied- and split-belt conditions to simulate and compare locomotion of intact and spinal cats. The major limitation of the present model is the lack of biomechanical elements simulating multi-joint limbs and muscles. Our study and analysis were specifically focused on the proposed organization and operation of spinal CPG circuits, including left-right interactions within these circuits, and their control by supraspinal drives and limb sensory feedback. Note that the term “supraspinal drive” in our model is used to represent supraspinal inputs providing both electrical and neuromodulator effects on spinal neurons increasing their excitability, which disappear after spinal transection. The other limitation of the present model is that it does not consider the possibility that afferent feedback can provide some constant level of excitation to the RG circuits after spinal transection, which can partly compensate for the lack of supraspinal drive and hence affect (shift) the timing of transitions between the considered regimes. We will consider this issue in the future. Another important limitation is that we did not consider possible mechanisms of plasticity following spinal transection. We know that the spinal connectome and sensorimotor interactions change after spinal cord injury [82,99]. The precise nature of these changes is not well understood. Nevertheless, we believe that changes (increase) in the gain of sensory feedback, which in the model results from eliminating presynaptic inhibition, in a real situation may include a considerable contribution from plastic changes during the recovery period. However, even with the account of possible plastic changes during the recovery period, the functional role of the increased gain of sensory feedback in spinal-transected animals remains the same independent of the exact mechanisms for this increase (release from presynaptic inhibition, plastic changes during recovery, currently unknown mechanisms, or any combination of the above).

Although the model was based on simplifications and some assumptions, it reproduced and provided explanations for experimental results, such as the main locomotor characteristics (cycle and phase durations) at different speeds and left-right speed differences during tied-belt and split-belt locomotion. We are currently applying our model to simulate cat locomotion following incomplete spinal cord injury (lateral hemisection) during tied-belt and split-belt locomotion based on our recent experimental data [100–102]. Using our model, this will allow us to remove supraspinal drives on the hemisected side and determine how the spinal network controls locomotion. We also plan to incorporate a full biomechanical model of the limbs to investigate the neuromechanical control of quadrupedal locomotion and its recovery following incomplete spinal cord injury [103,104].

Methods

Experimental data

For this modeling study, we used previously published data obtained from intact and spinal cats during tied-belt (equal left-right speeds) and split-belt (different left-right speeds) locomotion [11–13]. No new animals were used here. In those studies, all procedures were approved by the Animal Care Committee of the Université de Sherbrooke (Protocol 442-18) in accordance with policies and directives of the Canadian Council on Animal Care. Intact cats performed quadrupedal locomotion whereas spinal cats performed hindlimb-only locomotion with the forelimbs on a stationary platform. Intact and spinal cats performed tied-belt locomotion from 0.4 to 1.0 m/s and from 0.1 to 1.0 m/s, respectively, with 0.1 m/s increments. As stated earlier, intact cats cannot perform consistent quadrupedal tied-belt locomotion at or below 0.3 m/s. For split-belt locomotion, the slow belt (left) was 0.4 m/s while the fast belt (right) stepped at speeds of 0.5 to 1.0 m/s in 0.1 m/s increments.

Modeling formalism and model parameters

In our model, we considered spinal circuits as a network of interacting neural populations. Each population is described by an activity-based neuron model (and is sometimes called a *neuron* in the text), in which the dependent variable V represents an average population voltage and the output $f(V)$ ($0 \le f(V) \le 1$) represents the average or integrated population activity at the corresponding average voltage [13,30,31,105–107]. This description allows an explicit representation of ionic currents, specifically of the persistent (slowly inactivating) sodium current I_{NaP}(Rubin et al., 2009), which was proposed to be responsible for generating intrinsic bursting activity in the spinal cord [47,50,51,108–112]. Assuming that neurons within each population switch between silence and active spiking in a generally synchronized way, the dynamics of the average voltages are represented within a conductance-based framework used for a single neuron description, but without fast membrane currents responsible for spiking activity.

The dynamics of V for flexor and extensor half-centers, considered as I_{NaP}-dependent conditional bursters, is described by the following differential equation:

$$C \cdot \frac{dV}{dt} = -I_{\mathrm{NaP}} - I_{\mathrm{L}} - I_{\mathrm{SynE}} - I_{\mathrm{SynI}} \tag{1}$$

For other (non-bursting) populations, V is described as:

$$C \cdot \frac{dV}{dt} = -I_{\mathrm{L}} - I_{\mathrm{SynE}} - I_{\mathrm{SynI}} \tag{2}$$

The output function $f(V)$ transvers V to the integrated population output and is defined as follows:

$$f(V) = \begin{cases} 0, \text{ if } V < V_{\mathrm{thr}} \\ (V - V_{\mathrm{thr}})/(V_{\mathrm{max}} - V_{\mathrm{thr}}), \text{ if } V_{\mathrm{thr}} \le V < V_{\mathrm{max}} \\ 1, \text{ if } V \ge V_{\mathrm{max}} \end{cases} . \tag{3}$$

In Eqs. (1) and (2), C is the membrane capacitance, I_{NaP} is the persistent sodium current, $I_{\mathrm{L}}$is the leakage current, $I_{\mathrm{SynE}}$and I_{SynI}represent excitatory and inhibitory synaptic currents, respectively. The leakage current was described as:

$$I_{\mathrm{L}} = g_{\mathrm{L}} \cdot (V - E_{\mathrm{L}}), \tag{4}$$

where g_{L} is the leakage conductance and E_{L} represents the leakage reversal potential. The persistent sodium current in the flexor and extensor half-centers is described as:

$$I_{\mathrm{NaP}} = \bar{g}_{\mathrm{NaP}} \cdot m(V) \cdot h \cdot (V - E_{\mathrm{Na}}), \tag{5}$$

where $\bar{g}_{\mathrm{NaP}}$is the maximal conductance and $E_{\mathrm{Na}}$is the sodium reversal potential. Its voltage-dependent activation, $m(V)$, is instantaneous and its steady state is described as follows:

$$m(V) = m_{\infty}(V) = \left\{1 + \exp\left[\left(V - V_{1/2,m}\right)/k_m\right]\right\}^{-1} \tag{6}$$

The slow I_{NaP} inactivation was modeled by the following differential equation:

$$\tau_{\mathrm{h}}(V) \cdot \frac{dh}{dt} = h_{\infty}(V) - h, \tag{7}$$

where $h_{\infty}(V)$ represents inactivation steady state and $\tau_{\mathrm{h}}(V)$ is the inactivation time constant with maximal value τ_{max}.

$$h_{\infty}(V) = \left\{1 + \exp\left[\left(V - V_{1/2,h}\right)/k_h\right]\right\}^{-1}; \tag{8}$$

$$\tau_h(V) = \tau_{\mathrm{max}} / \cosh\left[\left(V - V_{1/2,\tau}\right)/k_{\tau}\right]. \tag{9}$$

In Eqs. (6), (8), and (9), $V_{1/2}$ and k represent half-voltage and slope of the corresponding variables *(m, h,* and *τ*). Excitatory and inhibitory synaptic currents (I_{SynE} and I_{SynI}) for population *i* were described by:

$$I_{\mathrm{SynE},i} = g_{\mathrm{SynE}} \cdot \left\{\sum_j\left[S\left(w_{ji}\right) \cdot f\left(V_j\right)\right] + D_i + SF_i^{\mathrm{E1}} + SF_i^{\mathrm{E2}}\right\} \cdot \left(V_i - E_{\mathrm{SynE}}\right); \tag{10}$$

$$I_{\mathrm{SynI},i} = g_{\mathrm{SynI}} \cdot \left\{\sum_j\left[S\left(-w_{ji}\right) \cdot f\left(V_j\right)\right]\right\} \cdot \left(V_i - E_{\mathrm{SynI}}\right), \tag{11}$$

where g_{SynE} and g_{SynI} are synaptic conductances and E_{SynE} and E_{SynI} are the reversal potentials of the excitatory and inhibitory synapses, respectively; w_{ji} is the synaptic weight from population *j* to population *i* ($w_{ji} > 0$ for excitatory connections and $w_{ji} < 0$ for inhibitory connections).

$$S(x) = \begin{cases} x, \text{ if } x \geq 0 \\ 0, \text{ if } x < 0 \end{cases}. \tag{12}$$

The weights of connections w_{ji} between populations for the main model (**Fig. 4**) are shown in **Table 1**. $D_i$in **Eq. (10)** for the main model (**Fig. 4**), represents the total excitatory drive to population *i*:

$$D_i = k_i^{\alpha ipsi} \cdot \alpha_{ipsi} + k_i^{\alpha contra} \cdot \alpha_{contra} + k_i^{\gamma} \cdot \gamma, \tag{13}$$

where α_{ipsi} and α_{contra} are the ipsi- and contralateral excitatory drives, respectively, which depending on the side, represent left α_L and right α_R drives, respectively; parameter γ represents the constant drive to the ipsilateral (left or right) extensor half-center; $k_i^{\alpha ipsi}$, $k_i^{\alpha contra}$, and k_i^{γ} define the weights of these drives to population *i*.

$SF_i^{\mathrm{E1}}$and SF_i^{E2} in **Eq. (10)** for the main model (**Fig. 4**), define the effect of ipsilateral sensory feedback E1 (or SF-E1, see **Fig. 4**) and E2 (or SF-E2), respectively, to population *i*. The gain of each feedback to population i ($k_i^{\mathrm{E1}}$and k_i^{E2}) is suppressed (reduced) by the ipsilateral drive α_{ipsi}

Weights of connections between neurons, w_{ji}	
Source neuron	Target neuron
F	InF (3.6), $V0_D$ (2.5), V2a (2.5), V3-F (1.0)
E	InE (3.6), V3-E (0.3)
InF	E (-8.0)
InE	F (-4.0)
V2a	$V0_V$ (2.5)
$V0_V$	c-Ini (2.5)
Ini	F (-2.0)
$V0_D$	c-F (–6)
V3-E	c-E (0.1), c-InE (1.5)
V3-F	c-F (0.1)
Weights of connections from drives to neurons: $k_i^{\alpha ipsi}, k_i^{\alpha contra}, k_i^{\gamma}$	
Source drive	Target neuron
$\alpha_{ipsi} \in [0; 1.0]$	F (0.45), InF(0.5), V3-E(0.1), V3-F(0.1)
$\alpha_{contra} \in [0; 1.0]$	F (0.15)
$\gamma = 1.0$	E (2.0)
Weights of connections from feedback sources to neurons: $k_i^{\mathrm{E1}}, k_i^{\mathrm{E2}}$	
Synaptic input from feedback	Target neuron
E1	F (1.4), V3-E (1.2)
E2	E (2.0)
Presynaptic inhibition of feedback connections by α_{ipsi}: $k_{PSi}^{\mathrm{E1}}, k_{PSi}^{\mathrm{E2}}$	
Synaptic input from feedback	Target feedback connection
E1	F (4.0), V3-E (0.5)
E2	E (2.0)

c – contralateral

Table 1.

Connection weights.

for the main model (**Fig. 4**), define the effect of ipsilateral sensory feedback E1 (or SF-E1, see **Fig. 4**) and E2 (or SF-E2), respectively, to population i. The gain of each feedback to population i (k_i^{E1} and k_i^{E2}) is suppressed (reduced) by the ipsilateral drive α_{ipsi} (α_L or α_R depending on the side):

$$SF_i^{\mathrm{E1}} = k_i^{\mathrm{E1}} \cdot \mathrm{E1}/(1 + k_{PSi}^{\mathrm{E1}} \cdot \alpha_{ipsi}); \tag{14}$$

$$SF_{\mathrm{i}}^{\mathrm{E2}} = k_i^{\mathrm{E2}} \cdot \mathrm{E2}/(1 + k_{PSi}^{\mathrm{E2}} \cdot \alpha_{ipsi}), \tag{15}$$

where k_{PSi}^{E1} and k_{PSi}^{E2} are the weights of presynaptic inhibition by α_{ipsi} of feedback inputs E1 and E2, respectively, to population i (see **Fig. 4A**).

E1 feedback (SF-E1) represents an increase of activity of length-dependent hip flexor afferents during limb extension. In our model, it is described as:

$$\tau_{\mathrm{E1}} \cdot \frac{d\mathrm{E1}}{dt} = [k_{\mathrm{E1}} \cdot \beta_{\mathrm{ipsi}}^{1.25} \cdot (t - t_0)] \cdot sign[f(V_{\mathrm{E}}) - thr] - \mathrm{E1}, \tag{16}$$

$$\text{where } \tau_{\mathrm{E1}} = \begin{cases} 0, \text{ if } \frac{d\mathrm{E1}}{dt} \geq 0; \\ \tau_1, \text{ if } \frac{d\mathrm{E1}}{dt} < 0; \end{cases}$$

$$sign[y] = \begin{cases} 1, \text{ if } y > 0; \\ 0, \text{ if } y \leq 0, \end{cases}$$

where k_{E1} is the parameter of E1 feedback ; β_{ipsi} is the parameter characterizing the speed of the ipsilateral treadmill belt (left β_L or right β_R), $f(V_{\mathrm{E}})$ is the output of the ipsilateral E half-center, *thr* is the threshold, t_0 is the starting time of the ipsilateral extensor burst and t is time (both in ms).

E2 feedback (SF-E2) that represents excitatory feedback from load-dependent afferents of limb extensor muscles to the extensor half-center during extension, described as follows:

$$\tau_{\mathrm{E2}} \cdot \frac{d\mathrm{E2}}{dt} = [E2o - k_{\mathrm{E2}} \cdot (t - t_0)] \cdot sign[f(V_{\mathrm{E}}) - thr] - \mathrm{E2}\,, \tag{17}$$

$$\text{where} \quad \tau_{\mathrm{E2}} = \begin{cases} \tau_2, \text{ if } \frac{d\mathrm{E2}}{dt} \geq 0; \\ 0, \text{ if } \frac{d\mathrm{E2}}{dt} < 0, \end{cases}$$

where $E2o$ and k_{E2} are the parameter of SF-E2.

Variables α_L and α_R characterize left and right supraspinal drives to RGs and define the frequency of locomotor oscillations. The variables $\beta_L$and β_R define the speeds of the left and right treadmill belts. Intermediate coefficients were used to make correspondence between α and β values and between their values and the speed of real treadmill belts in the experiments. For modeling of intact locomotion, we changed left and right α values to simulate “voluntary” locomotion in the intact model via a corresponding adjustment of supraspinal drives. For modeling of locomotion in the spinal-transected model, supraspinal drives were set to 0) and we only manipulated left and right β values simulating speeds of treadmill belts.

The following values of model parameters were used for the main model (**Fig. 4**): C = 20 pF; g_{L} = 4 nS for RG half-centers and g_{L} = 1 nS for all other neurons; $\bar{g}_{\mathrm{NaP}}$ = 4.4 nS; g_{synE} = g_{synI} = 1 nS; E_{L} = −60.0 mV; E_{Na} = 50.0 mV; E_{EsynE} = −10 mV; E_{EsynI} = −75 mV; V_{thr} = −50 mV; V_{max} = 0 mV; $V_{1/2,m}$ = −40.0 mV; k_m = −6 mV; $V_{1/2,h}$ = −45.0 mV; k_h = 4 mV; τ_{max} = 500 ms; $V_{1/2,\tau}$ = −45 mV; k_τ = 20 mV; $E2o$ = 2.5; $k_{\mathrm{E1}} = 3.12 \cdot 10^{-3}$ ms^{-1}; $k_{\mathrm{E2}} = 0.5 \cdot 10^{-3}$ ms^{-1}; τ_1 = 500 ms; τ_2 = 300 ms; *thr* =0.1. All connection weights for the main model (**Fig. 4**) are specified in **Table 1**.

In the models shown in **Fig. 3**: E_L = −64.0 mV; $w_1 = w_3$ = 3.6; w_2 = 3.0; w_4 = 5.0; Drive-E = 1.2. All other parameters are the same as in the main model, listed above.

Simulations, Data analysis and availability

All simulations were performed using the custom neural simulation package NSM 2.5.7. The simulation package was previously used for models of spinal circuits [7,27–29,32,50,51,111–113]. Differential equations were solved using the exponential Euler integration method with a step size of 0.1 ms. Simulation results were saved as ASCII files and represented the output functions for half-centers recorded with a precision of 0.1 ms.

The simulation results were processed using custom Matlab scripts (The Mathworks, Inc., Matlab 2023b). To assess model behavior, the activities of flexor and extensor half-centers were used to determine the onsets and offsets of flexor and extensor bursts and to calculate flexor and extensor phase durations and oscillation period. The timing of onsets and offsets of flexor and extensor bursts was determined at a threshold level of 0.05. The oscillation period was defined as the duration between two consecutive burst onsets in the left extensor half-center. The flexor and extensor phase durations and oscillation period were averaged over the duration of simulation for each value of the parameter α. Duration of individual simulations depended on the value of parameter α to robustly estimate average values of burst durations and oscillation period. For each α value, we omitted the first two transitional cycles to allow stabilization of model variables. Flexor and extensor phase durations and oscillation period were plotted against the parameters α and β.

The simulation package NSM 2.5.7, the model configuration file necessary to create and run simulations, and the custom Matlab scripts are available at *https://github.com/RybakLab/nsm*.

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Autor contributions

I.A.R. suggested the concept. A.F., B.I.P. and N.A.S. elaborated the details of the work. A.F. provided all experimental data. I.A.R. and N.A.S. developed the basic model. S.N.M. provided and supported simulation tools. N.A.S., I.A.R. and S.N.M. tuned model parameters. I.A.R., A.F. and B.I.P. interpreted the results. I.A.R., N.A.S., and A.F. prepared the figures. I.A.R. and A.F. drafted the manuscript. All authors discussed, reviewed, edited, and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

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Editors

Reviewing Editor
Jeffrey Smith
National Institute of Neurological Disorders and Stroke, Bethesda, United States of America

Senior Editor
Panayiota Poirazi
FORTH Institute of Molecular Biology and Biotechnology, Heraklion, Greece

Reviewer #1 (Public review):

Summary:

It is suggested that for each limb, the RG (rhythm generator) can operate in three different regimes: a non-oscillating state-machine regime and a flexor driven and a classical half-center oscillatory regime. This means that the field can move away from the old concept that there is only room for the classic half-center organization

Strengths:

A major benefit of the present paper is that a bridge was made between various CPG concepts ("a potential contradiction between the classical half-center and flexor-driven concepts of spinal RG operation"). Another important step forward is the proposal about the neural control of slow gait ("at slow speeds ({less than or equal to} 0.35 m/s), the spinal network operates in a state regime and requires external inputs for phase transitions, which can come from limb sensory feedback and/or volitional inputs (e.g. from the motor cortex").

Weaknesses:

Some references are missing

https://doi.org/10.7554/eLife.98841.2.sa3

Reviewer #2 (Public review):

Summary:

The biologically realistic model of the locomotor circuits developed by this group continues to define the state of the art for understanding spinal genesis of locomotion. Here the authors have achieved a new level of analysis of this model to generate surprising and potentially transformative new insights. They show that these circuits can operate in three very distinct states and that, in the intact spinal cord, these states come into successive operation as the speed of locomotion increases. Equally important, they show that in spinal injury, the model is "stuck" in the low-speed "state machine" behavior.

Strengths:

There are many strengths for the simulations results presented here. The model itself has been closely tuned to match a huge range of experimental data and this has a high degree of plausibility. The novel insight presented here, with the three different states, constitutes a truly major advance in the understanding of neural genesis of locomotion in spinal circuits. The authors systematically consider how the states of the model relate to presently available data from animal studies. Equally important, they provide a number of intriguing and testable predictions. It is likely that these insights are the most important achieved in the past

10 years. It is highly likely proposed multi-state behavior will have a transformative effect on this field.

Weaknesses:

I have no major weaknesses. A moderate concern is that the authors should consider some basic sensitivity analyses to determine if the 3-state behavior is especially sensitive to any of the major circuit parameters-e.g., connection strengths in the oscillators.

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Reviewer #3 (Public review):

General Comments

This work probes the control of walking in cats at different speeds and different states (split-belt and regular treadmill walking). Since the time of Sherrington there has been ongoing debate on this issue. The authors provide modeling data showing that they could reproduce data from cats walking on a specialized treadmill allowing for regular and split-belt walking. The data suggest that a non-oscillating state-machine regime best explains slow walking - where phase transitions are handled by external inputs into the spinal network. They then show at higher speeds a flexor-driven and then a classical half-center regime dominates. In spinal animals, it appears that a non-oscillating state-machine regime best explains the experimental data. The model is adapted from their previous work and raises interesting questions regarding the operation of spinal networks, that, at low speeds, challenge assumptions regarding central pattern generator function. This is an outstanding study which will be of general interest to the neuroscience community.

Strengths

The study has several strengths. Firstly the detailed model has been well established by the authors and provides details that relate to experimental data such as commissural interneurons (V0c and V0d), along with V3 and V2a interneuron data. Sensory input along with descending drive is also modelled and moreover the model reproduces many experimental data findings. Moreover, the idea that sensory feedback is more crucial at lower speeds, also is confirmed by presynaptic inhibition increasing with descending drive. The inclusion of experimental data from split-belt treadmills, and the ability of the model to reproduce findings here is a definite plus.

Weaknesses

Conceptually, this is a compelling study which provides interesting modeling data regarding the idea that the network can operate in different regimes, especially at lower speeds. The modelling data speaks for itself, but on the other hand, sensory feedback also provides generalized excitation of neurons which in turn project to the CPG. That is they are not considered part of the CPG proper. The authors have discussed this possibility in their revised paper.

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Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

It is suggested that for each limb the RG (rhythm generator) can operate in three different regimes: a non-oscillating state-machine regime, and in a flexor driven and a classical half-center oscillatory regime. This means that the field can move away from the old concept that there is only room for the classic half-center organization

Strengths:

A major benefit of the present paper is that a bridge was made between various CPG concepts ("a potential contradiction between the classical half-center and flexor-driven concepts of spinal RG operation"). Another important step forward is the proposal about the neural control of slow gait ("at slow speeds ({less than or equal to} 0.35 m/s), the spinal network operates in a state regime and requires external inputs for phase transitions, which can come from limb sensory feedback and/or volitional inputs (e.g. from the motor cortex").

Weaknesses:

Some references are missing

We thank the Reviewer for the thoughtful and constructive comments. We have added additional text to meet the specific Reviewer's recommendations and several references suggested by the Reviewer.

Reviewer #2 (Public Review):

Summary:

The biologically realistic model of the locomotor circuits developed by this group continues to define the state of the art for understanding spinal genesis of locomotion. Here the authors have achieved a new level of analysis of this model to generate surprising and potentially transformative new insights. They show that these circuits can operate in three very distinct states and that, in the intact cord, these states come into successive operation as the speed of locomotion increases. Equally important, they show that in spinal injury the model is "stuck" in the low speed "state machine" behavior.

Strengths:

There are many strengths for the simulation results presented here. The model itself has been closely tuned to match a huge range of experimental data and this has a high degree of plausibility. The novel insight presented here, with the three different states, constitutes a truly major advance in the understanding of neural genesis of locomotion in spinal circuits. The authors systematically consider how the states of the model relate to presently available data from animal studies. Equally important, they provide a number of intriguing and testable predictions. It is likely that these insights are the most important achieved in the past 10 years. It is highly likely proposed multi-state behavior will have a transformative effect on this field.

Weaknesses:

I have no major weaknesses. A moderate concern is that the authors should consider some basic sensitivity analyses to determine if the 3 state behavior is especially sensitive *to any of the major circuit parameters - e.g. connection strengths in the oscillators or?*

We thank the Reviewer for the thoughtful and constructive comments. The sensitivity analysis has been included as Supplemental file.

Reviewer #3 (Public Review):

Summary:

This work probes the control of walking in cats at different speeds and different states (split-belt and regular treadmill walking). Since the time of Sherrington there has been ongoing debate on this issue. The authors provide modeling data showing that they could reproduce data from cats walking on a specialized treadmill allowing for regular and split-belt walking. The data suggest that a non-oscillating state-machine regime best explains slow walking - where phase transitions are handled by external inputs into the spinal network. They then show at higher speeds a flexor-driven and then a classical halfcenter regime dominates. In spinal animals, it appears that a non-oscillating state-machine regime best explains the experimental data. The model is adapted from their previous work, and raises interesting questions regarding the operation of spinal networks, that, at low speeds, challenge assumptions regarding central pattern generator function. This is an interesting study. I have a few issues with the general validity of the treadmill data at low speeds, which I suspect can be clarified by the authors.

Strengths:

The study has several strengths. Firstly the detailed model has been well established by the authors and provides details that relate to experimental data such as commissural interneurons (V0c and V0d), along with V3 and V2a interneuron data. Sensory input along with descending drive is also modelled and moreover the model reproduces many experimental data findings. Moreover, the idea that sensory feedback is more crucial at lower speeds, also is confirmed by presynaptic inhibition increasing with descending drive. The inclusion of experimental data from split-belt treadmills, and the ability of the model to reproduce findings here is a definite plus.

Weaknesses:

Conceptually, this is a very useful study which provides interesting modeling data regarding the idea that the network can operate in different regimes, especially at lower speeds. The modelling data speaks for itself, but on the other hand, sensory feedback also provides generalized excitation of neurons which in turn project to the CPG. That is they are not considered part of the CPG proper. In these scenarios, it is possible that an appropriate excitatory drive could be provided to the network itself to move it beyond the state-machine state - into an oscillatory state. Did the authors consider that possibility? This is important since work using L-DOPA, for example, in cats or pharmacological activation of isolated spinal cord circuits, shows the CPG capable of producing locomotion without sensory or descending input.

We thank the Reviewer for the thoughtful and constructive comments. We have added additional texts, references, and discussed the issues raised by the Reviewer. Particularly, in section "Model limitations and future directions" we now admit that afferent feedback can provide some constant level excitation to the RG circuits after spinal transection which can partly compensate for the lack of supraspinal drive and hence affect (shift) the *timing of transitions between the considered regimes. We mentioned that this is one of the limitations of the present model. The potential effects of neuroactive drugs, like DOPA, on CPG circuits after spinal transection were left out because they are outside the scope of the present modeling studies.*

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

specific feedback to the authors:

Nevertheless, there are some minor points, worth considering.

Link to HUMAN DATA

Here the authors may be interested to know that human data supports their proposal. This is relevant since there is ample evidence for the operation of spinal CPG's in humans (Duysens and van de Crommert,1998). The present model predicts that the basic output of the CPG remains even at very slow speeds, thus leading to similarity in EMG output. This prediction fits the experimental data (den Otter AR, Geurts AC, Mulder T, Duysens J. Speed related changes in muscle activity from normal to very slow walking speeds. Gait Posture. 2004 Jun;19(3):270-8). To investigate whether the basic CPG output remains basically the same even at very slow speeds (as also predicted by the current model), humans walked slowly on a treadmill (speeds as slow as 0.28 m s−1). Results showed that the phasing of muscle activity remained relatively stable over walking speeds despite substantial changes in its amplitude. Some minor additions were seen, consistent with the increased demands of postural stability. Similar results were obtained in another study: Hof AL, Elzinga H, Grimmius W, Halbertsma JP. Speed dependence of averaged EMG profiles in walking. Gait Posture. 2002 Aug;16(1):78-86. doi:

10.1016/s0966-6362(01)00206-5. PMID: 12127190.

These authors wrote: "The finding that the EMG profiles of many muscles at a wide range of speeds can be represented by addition of few basic patterns is consistent with the notion of a central pattern generator (CPG) for human walking". The basic idea is that the same CPG can provide the motor program at slow and fast speeds but that the drive to the CPG differs. This difference is accentuated under some conditions in pathology, such as in Parkinson's Kinesia Paradoxa. It was argued that the paradox is not really a paradox but is explained as the CPGs are driven by different systems at slow and at fast speeds (Duysens J, Nonnekes J. Parkinson's Kinesia Paradoxa Is Not a Paradox. Mov Disord. 2021 May;36(5):1115-1118. doi: 10.1002/mds.28550. Epub 2021 Mar 3. PMID: 33656203.)

These ideas are well in line with the current proposal ("Based on our predictions, slow (conditionally exploratory) locomotion is not "automatic", but requires volitional (e.g. cortical) signals to trigger stepby-step phase transitions because the spinal network operates in a state-machine regime. In contrast, locomotion at moderate to high speeds (conditionally escape locomotion) occurs automatically under the control of spinal rhythm-generating circuits receiving supraspinal drives that define locomotor speed, unless voluntary modifications or precise stepping are required to navigate complex terrain").

As mentioned in the present paper, other examples exist from pathology ("...Another important implication of our results relates to the recovery of walking in movement disorders, where the recovered pattern is generally very slow. For example, in people with spinal cord injury, the recovered walking pattern is generally less than 0.1 m/s and completely lacks automaticity 77-79. Based on our predictions, because the spinal *locomotor network operates in a state-machine regime at these slow speeds, subjects need volition, additional external drive (e.g., epidural spinal cord stimulation) or to make use of limb sensory feedback by changing their posture to perform phase transitions"). As mentioned above, another example is provided by Parkinson's disease. The authors may also be interested in work on flexible generators in SCI: Danner SM, Hofstoetter US, Freundl B, Binder H, Mayr W, Rattay F, Minassian K. Human spinal locomotor control is based on flexibly organized burst generators. Brain. 2015 Mar;138(Pt 3):577-88. doi: 10.1093/brain/awu372. Epub 2015 Jan 12. PMID: 25582580; PMCID: PMC4408427.*

We thank the reviewer for these additional and interesting insights. We added a new paragraph in the Discussion to bolster the link with human data that includes references suggested by the Reviewer.

CHAIN OF REFLEXES

It reads: "... in opposition to the previously prevailing viewpoint of Charles Sherrington 21,22 that locomotion is generated through a chain of reflexes, i.e., critically depends on limb sensory feedback (reviewed in 23)." This is correct but incomplete. The reference cited (23: Stuart, D.G. and Hultborn, H, "Thomas Graham Brown (1882--1965), Anders Lundberg (1920-), and the neural control of stepping," Brain Res. Rev. 59(1), 74-95 (2008)) actually reads: "Despite the above findings, the doctrinaire position in the early 1900s was that the rhythm and pattern of hind limb stepping movements was attributable to sequential hind limb reflexes. According to Graham Brown (1911c) this viewpoint was largely due to the arguments of Sherrington and a Belgian physiologist, Maurice Philippson (1877-1938). Philippson studied stepping movements in chronically maintained spinal dogs, using techniques he had acquired in the Strasbourg laboratory of the distinguished German physiologist, Friedrich Goltz (1834-1902). He also analyzed kinematically moving pictures of dog locomotion, which had been sent to him by the renowned French physiologist, Etienne-Jules Marey (1830-1904). Philippson (1905) certainly presented arguments explaining his perception of how sequential spinal reflexes contributed to the four phases of the step cycle (see Fig. 1 in Clarac, 2008). In retrospect, it is likely that Graham Brown was correct in attributing to Philippson and Sherrington the then-prevailing viewpoint that reflexes controlled spinal stepping. It is puzzling, nonetheless, that far less was said then and even now about Philippson's belief that the spinal control was due to a combination of central and reflex mechanisms (Clarac, 2008),4,5 4 We are indebted to François Clarac for drawing to our attention Philippson's statement on p. 37 of his 1905 article that "Nos expériences prouvent d'une part que la moelle lombaire séparée du reste de l'axe cérébro-spinal est capable de produire les mouvements coordonnés dans les deux types de locomotion, trot et gallop. [Our experiments prove that one side of the spinal cord separated from the cerebro-spinal axis is able to produce coordinated movements in two types of locomotion, trot and gallop]." Then, on p. 39 Philippson (1905) states that "Nous voyons donc, en résumé que la coordination locomotrice est une fonction exclusivement médullaire, soutenue d'une part par des enchainements de réflexes directs et croisés, dont l'excitant est tantot le contact avec le sol, tantot le mouvement même du membre. [In summary, we see that locomotor coordination is an exclusive function of the spinal cord supported by a sequencing of direct and crossed reflexes, which are activated sometimes by contact with the ground and sometimes even by leg movement]. A coté de cette coordination basée sur des excitations périphériques, il y a une coordination centrale provenant des voies d'association intra-médullaires. [In conjunction with this peripherally excited coordination, there is a central coordination arising from intraspinal pathways]." (The English translations have also been kindly supplied by François Clarac.) Clearly, Philippson believed in both a central spinal and a reflex control of stepping! 5 In part 1 of his 1913/1916 review Graham Brown discussed Philippson's 1905 article in much

> *detail (pp. 345-350 in Graham Brown, 1913b). He concludes with the statement that "... Philippson die wesentlichen Factoren des Fortbewegungsaktes in das exterozeptive Nervensystem verlegt. Er nimmt an, dass die zyklischen Bewegungen automatisch durch äussere Reize erhalten werden, welche in sich selbst thythmisch als Folge der Reflexakte welche sie selbst erzeugen, wiederholt werden. [Philippson assigns the important factors of the act of locomotion to the exteroceptive nervous system. He assumes that the cyclic movements are automatically maintained by external stimuli which, by themselves, are rhythmically repeated as a consequence of the reflexive actions that they generate themselves]." (English translation kindly supplied by Wulfila Gronenberg). This interpretation clearly ignores Philippson's emphasis on a central spinal component in the control of stepping....). "*
>
> *Hence it is a simplification to give all credits to Sherrington and ignoring the role of Philippson concerning the chain of reflexes idea.*

We again thank the Reviewer for these additional and interesting insights. We added the Philippson (1905) and Clarac (2008) references. The important contribution of Philippson is now indicated.

> *GTO Ib feedback*
>
> *It reads: "This effect and the role of Ib feedback from extensor afferents has been demonstrated and described in many studies in cats during real and fictive locomotion 2,57-59."*
>
> *These citations are appropriate but it is surprising to see that the Hultborn contribution is limited to the Gossard reference while the even more important earlier reference to Conway et al is missing (Conway BA, Hultborn H, Kiehn O. Proprioceptive input resets central locomotor rhythm in the spinal cat. Exp Brain Res. 1987;68(3):643-56. doi: 10.1007/BF00249807. PMID: 3691733).*

Yes, the Conway et al. reference has been added.

> *Other species*
>
> *The authors may also look at other species. The flexible arrangement of the CPGs, as described in this article, is fully in line with work on other species, showing cpg networks capable to support gait, but also scratching, swimming ..etc (Berkowitz A, Hao ZZ. Partly shared spinal cord networks for locomotion and scratching. Integr Comp Biol. 2011 Dec;51(6):890-902. doi: 10.1093/icb/icr041. Epub 2011 Jun 22. PMID: 21700568. Berkowitz A, Roberts A, Soffe SR. Roles for multifunctional and specialized spinal interneurons during motor pattern generation in tadpoles, zebrafish larvae, and turtles. Front Behav Neurosci. 2010 Jun 28;4:36. doi: 10.3389/fnbeh.2010.00036. PMID: 20631847; PMCID: PMC2903196.)*
>
> *Similar ideas about flexible coupling can also be found in: Juvin L, Simmers J, Morin D. Locomotor rhythmogenesis in the isolated rat spinal cord: a phase-coupled set of symmetrical flexion extension oscillators. J Physiol. 2007 Aug 15;583(Pt 1):115-28. doi: 10.1113/jphysiol.2007.133413. Epub 2007 Jun 14. PMID: 17569737; PMCID: PMC2277226. Or zebrafish: Harris-Warrick RM. Neuromodulation and flexibility in Central Pattern Generator networks. Curr Opin Neurobiol. 2011 Oct;21(5):685-92. doi: 10.1016/j.conb.2011.05.011. Epub 2011 Jun 7. PMID: 21646013; PMCID: PMC3171584.*

We added a sentence in the Discussion along with supporting references.

> *Standing*

> *In the view of the present reviewer, the model could even be extended to standing in humans. It reads: "at slow speeds ({less than or equal to} 0.35 m/s), the spinal network operates in a state regime and requires external inputs"; similarly (personal experience) when going from sit to stand: as soon as weight is over support, extension is initiated and the body raises, as one would expect when the extensor center is activated by reinforcing load feedback, replacing GTO inhibition (Faist M, Hoefer C, Hodapp M, Dietz V, Berger W, Duysens J. In humans Ib facilitation depends on locomotion while suppression of Ib inhibition requires loading. Brain Res. 2006 Mar 3;1076(1):87-92. doi:*

Yes, we agree that the model could be extended to standing and the transition from standing to walking is particularly interesting. However, for this paper, we will keep the focus on locomotion over a range of speeds.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *The presentation is exceedingly well done and very clear.*
>
> *A moderate concern is that the authors do not make use of the capacity of computer simulations for sensitivity analyses. Perhaps these have been previously published? In any case, the question here is whether the 3 state behavior is especially sensitive to excitability of one of the main classes of neurons or a crucial set of connections.*

The sensitivity analysis has been made and included as Supplemental file.

> *Minor point. I have but two minor points. A bit more explanation should be provided for the use of the terms "state machine" to describe the lowest speed state. Perhaps this is a term from control theory? In any case, it is not clear why this is term is appropriate for a state in which the oscillator circuits are "stuck" in a constant output form and need to be "pushed" by sensory input.*

Yes, we now provide a definition in the Introduction.

> *Minor point: it is of course likely that neuromodulation of multiple types of spinal neurons occurs via inputs that activate G protein coupled receptors. These types of inputs are absent from the model, which is fine, but some sort of brief discussion should be included. One possibility is to note that the circuit achieves transitions between different states without the need for neuromodulatory inputs. This appears to me to be a very interesting and surprising insight.*

In section "Model limitations and future directions" in the Discussion**,** we now mention that the term "supraspinal drive" in our model is used to represent supraspinal inputs providing both electrical and neuromodulator effects on spinal neurons increasing their excitability, which disappear after spinal transection." We think that it is so far too early to simulate the exact effects of the descending neuromodulation, since there is almost no data on the effect of different modulators on specific types of spinal interneurons.

> ***Reviewer #3 (Recommendations For The Authors):***
>
> *Minor Comments*
>
> *Page numbers would be useful.*
>
> *Abstract*

> *Following spinal transection, the network can only operate in a state-machine regime. This is a bit strong since it applies to computational data. Clarify this statement.*

We agree. Sentence has been changed to: “Following spinal transection, the model predicts that the spinal network can only operate in the state-machine regime.”

> *Introduction*
>
> *Intro - "This is somewhat surprising...". It gives the impression that spinal cats are autonomously stable on the belt. They are stabilized by the experimenter.*

The text has been changed to: “This is somewhat surprising because intact and spinal cats rely on different control mechanisms. Intact cats walking freely on a treadmill engage vision for orientation in space and their supraspinal structures process visual information and send inputs to the spinal cord to control locomotion on a treadmill that maintains a fixed position of the animal relative to the external space. Spinal cats, whose position on the treadmill relative to the external space is fixed by an experimenter, can only use sensory feedback from the hindlimbs to adjust locomotion to the treadmill speed.”

> *"Cannot consistently perform treadmill locomotion" - likely a context-dependent result. Certainly, cats can do this easily off a treadmill - stalking, for example. Perhaps somewhere, mention that treadmill locomotion is not entirely similar to overground locomotion.*

We completely agree. Stalking is an excellent example showing that during overground locomotion slow movements (and related phase transitions) can be controlled by additional voluntary commands from supraspinal structures, which differs from simple treadmill locomotion, performing out of specific goalor task-dependent contexts. Based on this, we suggest a difference between a relatively slow (exploratory-type, including stalking) and relatively fast (escape-type) overground locomotion. We added the following sentence to the introduction:” This is evidently context dependent and specific for the treadmill locomotion as cats, humans and other animals can voluntarily decide to perform consistent overground locomotion at slow speeds.”

> *The authors introduce the concept of the state machine regime. In my opinion, this could use some more explanation and citations to the literature. Was it a term coined by the authors, or is there literature reinforcing this point?*

This is a computer science and automata theory term that has already been used in descriptions of locomotion (see our references in the 2nd paragraph of Discussion). We added a definition and corresponding references in the Introduction.

> *In terms of sensory feedback, particularly group II input, it would be interesting to calculate if the conduction delay to the spinal cord at higher speeds would have a certain cutoff point at which it would no longer be timed effectively for phase transitions. This could reinforce your point.*

This is an interesting proposition but it is unlikely to be a factor over the range of speeds that we investigated (0.1 to 1.0 m/s). Assuming that group II afferents transmit their signals to spinal circuits at a latency of 10-20 ms, this is more than enough time to affect phase transitions, even at the highest speed considered. This might be a factor at very high speeds (e.g. galloping) or in small animals with high stepping frequencies.

> *Results.*

> *The assertion that intact cats are inconsistent in terms of walking at slow speeds needs to be bolstered. For example, if a raised platform were built for a tray of food, would the intact cat consistently walk at slower speeds and eat? I suspect so. By the same token, would they walk slowly during bipedal walking? It is pretty easy to check this. Also, reports from the literature show differential effects of runway versus treadmill gait analysis, specifically when afferent input is removed.*

The Reviewer is correct that raising a platform for a food tray or even having intact cats walk with their hindlimbs only (with forelimbs on a stationary platform) may allow for consistent stepping at slow speeds (0.1 – 0.3 m/s). However, this effectively removes voluntary control of locomotion and makes the pattern more automatic (spinal + limb sensory feedback). These examples provide additional specific contexts, and we have already mentioned (see above) that slow locomotion of intact cat is context dependent.

> *"We believe that intact animals walking on a treadmill..." Citations for this? Certainly, this is not a new point.*

No, this is not new. We changed the sentence and added a reference to the statement: "Intact animals walking on a treadmill use visual cues and supraspinal signals to adjust their speed and maintain a fixed position relative to the external space with reference to Salinas et al. (Salinas, M.M., Wilken, J M, and Dingwell, J B, "How humans use visual optic flow to regulate stepping during walking," Gait. Posture. 57, 15-20, 2017).

> *The presentation of the results is somewhat disjointed. The intact data is presented for tied and splitbelt results, but this is not addressed explicitly until figure 4. Would it not be better to create a figure incorporating both intact and modelling data and present the intact data where appropriate?*

We tried to do this initially, but this way required changing the style of the whole paper and we decided against this idea. Therefore, we prefer to keep the presentation of results as it is now.

> *Regarding the role of sensory feedback being especially important at low speeds, it is interesting that egr3+ mice (lacking spindle input) show an inability to walk at high speeds >40 cm/s but can walk at lower speeds (up to 7 cm/s) (Takeoka et al 2014). Similar findings were found with a lesion affecting Group I afferents in general (Takeoka and Arber 2019). Also, Grillner and colleagues show that cats can produce fictive locomotion in the absence of sensory input.*

In the Takeoka experiments it is difficult to assess the effect of removing somatosensory feedback because animals can simply decide to not step at higher speeds to avoid injury. Their mice deprived of somatosensory feedback can walk at slow speeds, likely thanks to voluntary commands, and cannot do so at higher speeds because (1) maybe somatosensory feedback is indeed necessary and/or (2) because they feel threatened because of impaired posture and poor control in general. In other words, they choose to not walk at faster speeds to avoid injury.

Fictive locomotion by definition is without phasic somatosensory feedback as the animals are curarized or studies are performed in isolated spinal cord preparations. Depending on the preparation, pharmacology or brainstem stimulation is required to evoke fictive locomotion. If animals are deafferented, pharmacology or brainstem stimulation are required to induce fictive locomotion to offset the loss of spinal neuronal excitability provided by primary afferents. At the same time, our preliminary analysis of old fictive locomotion data in the University of Manitoba Spinal Cord center (Drs. Markin and Rybak had an official access to these data base during our collaboration with Dr. David McCrea) has shown that the frequency of stable fictive locomotion in cats usually exceeded 0.6 - 0.7 Hz, which approximately corresponds to the speed above 0.3 - 0.4 m/s. These data and estimation are just approximate; they have not been statistically analyzed and published and hence have not been included in our paper.

> *Discussion. The statement that sensory feedback is required for animals to locomote may need to be qualified. Animals need some sensory feedback to locomote is perhaps better. For example, lesion studies by Rossignol in the early 2000s showed that cutaneous feedback from the paw was seemingly quite critical (in spinal cats). Also, see previous comments above.*

We changed this to: "… requires some sensory feedback to locomote, …"

> *Figures*
>
> *Figure 1C. This figure is somewhat confusing. If intact cats do not walk (arrow), how are the data for swing and stance computed? Also raw traces would be useful to indicate that there is variability. Also, while duration is useful, would you not want to illustrate the co-efficient of variation as well as another way to show that the stepping pattern was inconsistent?*

This is probably a misunderstanding. The left panel of Fig. 1C superimposes data of intact cats from panel A (with speed range from 0.4 m/s to 1.0 m/s) and data from spinal cats from panel B (with speed range from 0.1 m/s and 1.0 m/s). Therefore, the left part of this left panel 1C (with speed range from 0.1 m/s to 0.4 m/s (pointed out by the arrow) corresponds only to spinal cats (not to intact cats). The standard deviations of all measurements are shown. All these figures were reproduced from the previous publications. We did not apply new statistical analysis to these previously published data/figures.

> *Figure 4. 'All supraspinal drives (and their suppression of sensory feedback) are eliminated from the schematic shown in A. ' However, it is labelled 'brainstem drives,' which is confusing. Moreover, many of the abbreviations are confusing. Do you need I-SF-E1 in the figure, or could you call it 'Feedback 1' and then refer to I-SF-E1 in the legend? The same goes for βr, etc. Can they move to the legend?*

In the intact model (Fig. 4A), we have supraspinal drives (αL and αR, and γL and γR), some of which provide presynaptic inhibition of sensory feedback (SF-E1 and SF-E2) as shown in Fig. 4A. In spinaltransected model (Fig. 4B), the above brainstem drives and their effects (presynaptic inhibition) on both feedback types are eliminated (therefore, there is no label "Brainstem drives in Fig. 4B). Also, we do not see a strong reason to change the feedback names, since they are explained in the text.

> *I appreciate the detail of these figures, but they are difficult to conceptualize. They are useful in the context of 3C. Perhaps move this figure to supplementary and then show the proposed schematics for the system operating at slow, medium, and fast speeds in a replacement figure?*

We apologize for the resistance, but we would like to keep the current presentation.

> *There is a lack of raw data (models or experimental) data reinforcing the figures. I would add these to all figures, which would nicely complement the graphs.*

These raw data can be found in the cited manuscripts. It would be the same figures.

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